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Systems, methods, and devices for electronic spectrum management for identifying open space

Abstract

Systems, methods, and apparatus are provided for automated identification of open space in a wireless communications spectrum, by identifying sources of signal emission in the spectrum by automatically detecting signals, analyzing signals, comparing signal data to historical and reference data, creating corresponding signal profiles, and determining information about the open space based upon the measured and analyzed data in near real-time.

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continuation parent-doc US 14504819 20141002 US 9635573 20170425 child-doc US 15492475
continuation parent-doc US 14273157 20140508 US 8873422 20141028 child-doc US 14329838
continuation parent-doc US 14086871 20131121 US 8750156 20140610 child-doc US 14273157
continuation parent-doc US 13912893 20130607 US 9078162 20150707 child-doc US 14082916
continuation parent-doc US 13912683 20130607 US 9288683 20160315 child-doc US 14082873
continuation parent-doc US 13913013 20130607 US 9622041 20170411 child-doc US 14082930
continuation-in-part parent-doc US 14082873 20131118 US 8805291 20140812 child-doc US
14086871
continuation-in-part parent-doc US 14082916 20131118 US 8780968 20140705 child-doc US
14086871
continuation-in-part parent-doc US 14082930 20131118 US 8824536 20140902 child-doc US
14086871
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4215345	12/1979	Robert et al.	N/A	N/A
4400700	12/1982	Rittenbach	N/A	N/A
4453137	12/1983	Rittenbach	N/A	N/A
4501020	12/1984	Wakeman	N/A	N/A
4581769	12/1985	Grimsley et al.	N/A	N/A
4638493	12/1986	Bishop et al.	N/A	N/A

4794325	12/1987	Britton et al.	N/A	N/A
4928106	12/1989	Ashjaee et al.	N/A	N/A
5103402	12/1991	Morton et al.	N/A	N/A
5134407	12/1991	Lorenz et al.	N/A	N/A
5144642	12/1991	Weinberg et al.	N/A	N/A
5166664	12/1991	Fish	N/A	N/A
5230087	12/1992	Meyer et al.	N/A	N/A
5293170	12/1993	Lorenz et al.	N/A	N/A
5343212	12/1993	Rose et al.	N/A	N/A
5393713	12/1994	Schwob	N/A	N/A
5416593	12/1994	Vercruysse	N/A	N/A
5448309	12/1994	Won	N/A	N/A
5506864	12/1995	Schilling	N/A	N/A
5513385	12/1995	Tanaka	N/A	N/A
5548809	12/1995	Lemson	N/A	N/A
5570099	12/1995	DesJardins	N/A	N/A
5589835	12/1995	Gildea et al.	N/A	N/A
5612703	12/1996	Mallinckrodt	N/A	N/A
5642732	12/1996	Wang	N/A	N/A
5831874	12/1997	Boone et al.	N/A	N/A
5835857	12/1997	Otten	N/A	N/A
5838906	12/1997	Doyle et al.	N/A	N/A
5846208	12/1997	Pichlmayr et al.	N/A	N/A
5856803	12/1998	Pevler	N/A	N/A
5936575	12/1998	Azzarelli et al.	N/A	N/A
6018312	12/1999	Haworth	N/A	N/A
6039692	12/1999	Kristoffersen	N/A	N/A
6085090	12/1999	Yee et al.	N/A	N/A
6115580	12/1999	Chuprun et al.	N/A	N/A
6134445	12/1999	Gould et al.	N/A	N/A
6144336	12/1999	Preston et al.	N/A	N/A
6157619	12/1999	Ozluturk et al.	N/A	N/A
6160511	12/1999	Pfeil et al.	N/A	N/A
6167277	12/1999	Kawamoto	N/A	N/A
6185309	12/2000	Attias	N/A	N/A
6188715	12/2000	Partyka	N/A	N/A
6191731	12/2000	McBurney et al.	N/A	N/A
6198414	12/2000	McPherson et al.	N/A	N/A
6249252	12/2000	Dupray	N/A	N/A
6286021	12/2000	Tran et al.	N/A	N/A
6296612	12/2000	Mo et al.	N/A	N/A
6304760	12/2000	Thomson et al.	N/A	N/A
6314366	12/2000	Farmakis et al.	N/A	N/A
6339396	12/2001	Mayersak	N/A	N/A
6400647	12/2001	Huntress	N/A	N/A
6418131	12/2001	Snelling et al.	N/A	N/A
6430418	12/2001	Nivens et al.	N/A	N/A
6433671	12/2001	Nysen	N/A	N/A
6490318	12/2001	Larsson et al.	N/A	N/A
6492945	12/2001	Counselman, III et al.	N/A	N/A

6512788	12/2002	Kuhn et al.	N/A	N/A
6628231	12/2002	Mayersak	N/A	N/A
6677895	12/2003	Holt	N/A	N/A
6697439	12/2003	Trivedi et al.	N/A	N/A
6707910	12/2003	Valve et al.	N/A	N/A
6711404	12/2003	Arpee et al.	N/A	N/A
6741595	12/2003	Maher et al.	N/A	N/A
6771957	12/2003	Chitrapu	N/A	N/A
6785321	12/2003	Yang et al.	N/A	N/A
6850557	12/2004	Gronemeyer	N/A	N/A
6850735	12/2004	Sugar et al.	N/A	N/A
6859831	12/2004	Gelvin et al.	N/A	N/A
6861982	12/2004	Forstrom et al.	N/A	N/A
6876326	12/2004	Martorana	N/A	N/A
6898197	12/2004	Lavean	N/A	N/A
6898235	12/2004	Carlin et al.	N/A	N/A
6904269	12/2004	Deshpande et al.	N/A	N/A
6985437	12/2005	Vogel	N/A	N/A
6991514	12/2005	Meloni et al.	N/A	N/A
7035593	12/2005	Miller et al.	N/A	N/A
7043207	12/2005	Miyazaki	N/A	N/A
7049965	12/2005	Kelliher et al.	N/A	N/A
7110756	12/2005	Diener	N/A	N/A
7116943	12/2005	Sugar et al.	N/A	N/A
7146176	12/2005	Mchenry	N/A	N/A
7151790	12/2005	Patenaude et al.	N/A	N/A
7151938	12/2005	Weigand	N/A	N/A
7152025	12/2005	Lusky et al.	N/A	N/A
7162207	12/2006	Kursula et al.	N/A	N/A
7171161	12/2006	Miller	N/A	N/A
7187326	12/2006	Beadle et al.	N/A	N/A
7206350	12/2006	Korobkov et al.	N/A	N/A
7215716	12/2006	Smith	N/A	N/A
7254191	12/2006	Sugar et al.	N/A	N/A
7269151	12/2006	Diener et al.	N/A	N/A
7289733	12/2006	He	N/A	N/A
7292656	12/2006	Kloper et al.	N/A	N/A
7298327	12/2006	Dupray et al.	N/A	N/A
7340375	12/2007	Patenaud et al.	N/A	N/A
7366463	12/2007	Archer et al.	N/A	N/A
7408907	12/2007	Diener	N/A	N/A
7424268	12/2007	Diener et al.	N/A	N/A
7428270	12/2007	Dubuc et al.	N/A	N/A
7430254	12/2007	Anderson	N/A	N/A
7459898	12/2007	Woodings	N/A	N/A
7466960	12/2007	Sugar	N/A	N/A
7471683	12/2007	Maher, III et al.	N/A	N/A
7522917	12/2008	Purdy, Jr. et al.	N/A	N/A
7555262	12/2008	Brenner	N/A	N/A
7564816	12/2008	Mchenry et al.	N/A	N/A

7595754	12/2008	Mehta	N/A	N/A
7606335	12/2008	Kloper et al.	N/A	N/A
7606597	12/2008	Weigand	N/A	N/A
7620396	12/2008	Floam et al.	N/A	N/A
7653020	12/2009	Roberts	N/A	N/A
7676192	12/2009	Wilson	N/A	N/A
7692532	12/2009	Fischer et al.	N/A	N/A
7692573	12/2009	Funk	N/A	N/A
7702044	12/2009	Nallapureddy et al.	N/A	N/A
7725110	12/2009	Weigand	N/A	N/A
7728755	12/2009	Jocic	N/A	N/A
7801490	12/2009	Scherzer	N/A	N/A
7813742	12/2009	Mitchell	N/A	N/A
7835319	12/2009	Sugar	N/A	N/A
7865140	12/2010	Levien et al.	N/A	N/A
7893875	12/2010	Smith	N/A	N/A
7929508	12/2010	Yucek et al.	N/A	N/A
7933344	12/2010	Hassan et al.	N/A	N/A
7945215	12/2010	Tang	N/A	N/A
7953549	12/2010	Graham et al.	N/A	N/A
7965641	12/2010	Ben Letaief et al.	N/A	N/A
8001901	12/2010	Bass	N/A	N/A
8006195	12/2010	Woodings et al.	N/A	N/A
8023957	12/2010	Weigand	N/A	N/A
8026846	12/2010	Mcfadden et al.	N/A	N/A
8027249	12/2010	Mchenry et al.	N/A	N/A
8027690	12/2010	Shellhammer	N/A	N/A
8045654	12/2010	Anderson	N/A	N/A
8045660	12/2010	Gupta	N/A	N/A
8055204	12/2010	Livsics et al.	N/A	N/A
8059694	12/2010	Junell et al.	N/A	N/A
8060017	12/2010	Schlicht et al.	N/A	N/A
8060035	12/2010	Haykin	N/A	N/A
8060104	12/2010	Chaudhri et al.	N/A	N/A
8064840	12/2010	McHenry et al.	N/A	N/A
8077662	12/2010	Srinivasan et al.	N/A	N/A
RE43066	12/2011	McHenry	N/A	N/A
8094610	12/2011	Wang et al.	N/A	N/A
8107391	12/2011	Wu et al.	N/A	N/A
8125213	12/2011	Goguillon et al.	N/A	N/A
8131239	12/2011	Walker et al.	N/A	N/A
8134493	12/2011	Noble et al.	N/A	N/A
8151311	12/2011	Huffman et al.	N/A	N/A
8155039	12/2011	Wu et al.	N/A	N/A
8155649	12/2011	McHenry et al.	N/A	N/A
8160839	12/2011	Woodings et al.	N/A	N/A
8170577	12/2011	Singh	N/A	N/A
8175539	12/2011	Diener et al.	N/A	N/A
8184653	12/2011	Dain et al.	N/A	N/A
8193981	12/2011	Hwang et al.	N/A	N/A

8213868	12/2011	Du et al.	N/A	N/A
8224254	12/2011	Haykin	N/A	N/A
8229368	12/2011	Immendorf et al.	N/A	N/A
8233928	12/2011	Stanforth et al.	N/A	N/A
8238247	12/2011	Wu et al.	N/A	N/A
8249028	12/2011	Porras et al.	N/A	N/A
8249631	12/2011	Sawai	N/A	N/A
8260207	12/2011	Srinivasan et al.	N/A	N/A
8265684	12/2011	Sawai	N/A	N/A
8279786	12/2011	Smith et al.	N/A	N/A
8280433	12/2011	Quinn et al.	N/A	N/A
8289907	12/2011	Seidel et al.	N/A	N/A
8290503	12/2011	Sadek et al.	N/A	N/A
8295859	12/2011	Yarkan et al.	N/A	N/A
8295877	12/2011	Hui et al.	N/A	N/A
8301075	12/2011	Sherman et al.	N/A	N/A
8305215	12/2011	Markhovsky et al.	N/A	N/A
8311483	12/2011	Tillman et al.	N/A	N/A
8311509	12/2011	Feher	N/A	N/A
8315571	12/2011	Lindoff et al.	N/A	N/A
8320910	12/2011	Bobier	N/A	N/A
8326240	12/2011	Kadambe et al.	N/A	N/A
8326309	12/2011	Mody et al.	N/A	N/A
8326313	12/2011	McHenry et al.	N/A	N/A
8335204	12/2011	Samarasooriya et al.	N/A	N/A
8346273	12/2012	Weigand	N/A	N/A
8350970	12/2012	Birkett et al.	N/A	N/A
8352223	12/2012	Anthony et al.	N/A	N/A
8358723	12/2012	Hamkins et al.	N/A	N/A
8364188	12/2012	Srinivasan et al.	N/A	N/A
8369305	12/2012	Diener et al.	N/A	N/A
8373759	12/2012	Samarasooriya et al.	N/A	N/A
8391794	12/2012	Sawai et al.	N/A	N/A
8391796	12/2012	Srinivasan et al.	N/A	N/A
8401564	12/2012	Singh	N/A	N/A
8406776	12/2012	Jallon	N/A	N/A
8406780	12/2012	Mueck	N/A	N/A
RE44142	12/2012	Wilson	N/A	N/A
8421676	12/2012	Moshfeghi	N/A	N/A
8422453	12/2012	Abedi	N/A	N/A
8422958	12/2012	Du et al.	N/A	N/A
RE44237	12/2012	Mchenry	N/A	N/A
8437700	12/2012	Mody et al.	N/A	N/A
8442445	12/2012	Mody et al.	N/A	N/A
8447237	12/2012	Reial et al.	N/A	N/A
8451751	12/2012	Challapali et al.	N/A	N/A
8463195	12/2012	Shellhammer	N/A	N/A
8467353	12/2012	Proctor	N/A	N/A
8467996	12/2012	Perez	702/70	G01J 3/0264

8483155	12/2012	Banerjea et al.	N/A	N/A
8494464	12/2012	Kadambe et al.	N/A	N/A
8503955	12/2012	Kang et al.	N/A	N/A
8504087	12/2012	Stanforth et al.	N/A	N/A
8514729	12/2012	Blackwell	N/A	N/A
8515473	12/2012	Mody et al.	N/A	N/A
8520606	12/2012	Cleveland	N/A	N/A
RE44492	12/2012	Mchenry	N/A	N/A
8526974	12/2012	Olsson et al.	N/A	N/A
8532686	12/2012	Schmidt et al.	N/A	N/A
8538339	12/2012	Hu et al.	N/A	N/A
8548521	12/2012	Hui et al.	N/A	N/A
8554264	12/2012	Gibbons et al.	N/A	N/A
8559301	12/2012	Mchenry et al.	N/A	N/A
8565811	12/2012	Tan et al.	N/A	N/A
8599024	12/2012	Bloy	N/A	N/A
8718838	12/2013	Kokkeby et al.	N/A	N/A
8761051	12/2013	Brisebois et al.	N/A	N/A
8773966	12/2013	Petrovic et al.	N/A	N/A
8780968	12/2013	Garcia et al.	N/A	N/A
8792901	12/2013	Schmidt	455/67.11	H04W 16/14
8798548	12/2013	Carbajal	N/A	N/A
8805291	12/2013	Garcia et al.	N/A	N/A
8818283	12/2013	McHenry et al.	N/A	N/A
8824536	12/2013	Garcia et al.	N/A	N/A
8843155	12/2013	Burton et al.	N/A	N/A
8941491	12/2014	Polk et al.	N/A	N/A
8977212	12/2014	Carbajal	N/A	N/A
9007262	12/2014	Witzgall	N/A	N/A
9008587	12/2014	Carbajal	N/A	N/A
9078162	12/2014	Garcia et al.	N/A	N/A
9143968	12/2014	Manku et al.	N/A	N/A
9185591	12/2014	Carbajal	N/A	N/A
9229102	12/2015	Wright et al.	N/A	N/A
9245378	12/2015	Villagomez et al.	N/A	N/A
9288683	12/2015	Garcia et al.	N/A	N/A
9356727	12/2015	Immendorf et al.	N/A	N/A
9397619	12/2015	Lozhkin	N/A	N/A
9412278	12/2015	Gong et al.	N/A	N/A
9413574	12/2015	Timofeev et al.	N/A	N/A
9414237	12/2015	Garcia et al.	N/A	N/A
9439078	12/2015	Menon et al.	N/A	N/A
9466881	12/2015	Berry et al.	N/A	N/A
9529360	12/2015	Melamed et al.	N/A	N/A
9537586	12/2016	Carbajal	N/A	N/A
9538040	12/2016	Goergen et al.	N/A	N/A
9572055	12/2016	Immendorf et al.	N/A	N/A
9635669	12/2016	Gormley et al.	N/A	N/A
9658341	12/2016	Mathews et al.	N/A	N/A

9674684	12/2016	Mendelson	N/A	N/A
9674836	12/2016	Gormley et al.	N/A	N/A
9686789	12/2016	Gormley et al.	N/A	N/A
9715009	12/2016	Parker et al.	N/A	N/A
9749069	12/2016	Garcia et al.	N/A	N/A
9755972	12/2016	Mao et al.	N/A	N/A
9767699	12/2016	Borghese et al.	N/A	N/A
9769834	12/2016	Immendorf et al.	N/A	N/A
9805273	12/2016	Seeber et al.	N/A	N/A
9819441	12/2016	Immendorf et al.	N/A	N/A
9858947	12/2017	Hearing et al.	N/A	N/A
9862489	12/2017	Weinstein et al.	N/A	N/A
9923700	12/2017	Gormley et al.	N/A	N/A
9942775	12/2017	Yun et al.	N/A	N/A
9973278	12/2017	Wang et al.	N/A	N/A
9989633	12/2017	Pandey et al.	N/A	N/A
9998243	12/2017	Garcia et al.	N/A	N/A
10027429	12/2017	Kiannejad	N/A	N/A
10104559	12/2017	Immendorf et al.	N/A	N/A
10157548	12/2017	Priest	N/A	N/A
10194324	12/2018	Yun et al.	N/A	N/A
10198955	12/2018	Boyd et al.	N/A	N/A
10227429	12/2018	Watanabe et al.	N/A	N/A
10235523	12/2018	Keller, III et al.	N/A	N/A
10241140	12/2018	Moinuddin	N/A	N/A
10251242	12/2018	Rosen et al.	N/A	N/A
10281570	12/2018	Parker et al.	N/A	N/A
10389616	12/2018	Ryan et al.	N/A	N/A
10393784	12/2018	Logan et al.	N/A	N/A
10408936	12/2018	Van Voorst	N/A	N/A
10459020	12/2018	Dzierwa et al.	N/A	N/A
10540905	12/2019	Bohanan et al.	N/A	N/A
10552738	12/2019	Holt et al.	N/A	N/A
10587352	12/2019	Kiannejad	N/A	N/A
10594034	12/2019	Tran et al.	N/A	N/A
10613209	12/2019	Emami et al.	N/A	N/A
10642813	12/2019	Lazier et al.	N/A	N/A
10698076	12/2019	Jones et al.	N/A	N/A
10700721	12/2019	Ayala et al.	N/A	N/A
10701574	12/2019	Gormley et al.	N/A	N/A
10764718	12/2019	Boettcher et al.	N/A	N/A
10784974	12/2019	Menon	N/A	N/A
10811771	12/2019	Tran et al.	N/A	N/A
10907940	12/2020	Parker et al.	N/A	N/A
10916845	12/2020	Tran et al.	N/A	N/A
10917797	12/2020	Menon et al.	N/A	N/A
11012340	12/2020	Ryan et al.	N/A	N/A
11035929	12/2020	Parker et al.	N/A	N/A
11063653	12/2020	Ottersten et al.	N/A	N/A
11190233	12/2020	Lo et al.	N/A	N/A

11223431	12/2021	Garcia et al.	N/A	N/A
11265652	12/2021	Kallai et al.	N/A	N/A
11321282	12/2021	Tran	N/A	N/A
11334807	12/2021	O'Shea et al.	N/A	N/A
11336011	12/2021	Tran et al.	N/A	N/A
11516071	12/2021	Karapantelakis et al.	N/A	N/A
11637641	12/2022	Garcia et al.	N/A	N/A
11663992	12/2022	Canberk et al.	N/A	N/A
11671839	12/2022	Guo et al.	N/A	N/A
11700304	12/2022	Brown, Jr. et al.	N/A	N/A
11757185	12/2022	Tran et al.	N/A	N/A
11777783	12/2022	Meirosu et al.	N/A	N/A
11791913	12/2022	Garcia et al.	N/A	N/A
11871103	12/2023	Kleinbeck	N/A	N/A
11874674	12/2023	Groden	N/A	G08G 5/55
11880888	12/2023	Gold et al.	N/A	N/A
11889351	12/2023	Tagg	N/A	N/A
11910305	12/2023	Buyukdura	N/A	N/A
12095518	12/2023	Garcia et al.	N/A	N/A
12156037	12/2023	Montalvo	N/A	N/A
2001/0000959	12/2000	Campana et al.	N/A	N/A
2001/0005423	12/2000	Rhoads	N/A	N/A
2001/0016503	12/2000	Kang	N/A	N/A
2001/0020220	12/2000	Kurosawa	N/A	N/A
2002/0044082	12/2001	Woodington et al.	N/A	N/A
2002/0070889	12/2001	Griffin et al.	N/A	N/A
2002/0072331	12/2001	Fischer et al.	N/A	N/A
2002/0097184	12/2001	Mayersak	N/A	N/A
2002/0119754	12/2001	Wakutsu et al.	N/A	N/A
2002/0161775	12/2001	Lasensky et al.	N/A	N/A
2002/0173341	12/2001	Abdelmonem et al.	N/A	N/A
2003/0013454	12/2002	Hunzinger	N/A	N/A
2003/0040277	12/2002	Deats	N/A	N/A
2003/0083091	12/2002	Nuutinen et al.	N/A	N/A
2003/0087648	12/2002	Mezhvinsky et al.	N/A	N/A
2003/0104831	12/2002	Razavilar et al.	N/A	N/A
2003/0144601	12/2002	Prichep	N/A	N/A
2003/0145008	12/2002	Burrell	N/A	N/A
2003/0145328	12/2002	Rabinowitz et al.	N/A	N/A
2003/0198304	12/2002	Sugar et al.	N/A	N/A
2003/0206640	12/2002	Malvar et al.	N/A	N/A
2003/0224801	12/2002	Lovberg et al.	N/A	N/A
2003/0232612	12/2002	Richards et al.	N/A	N/A
2004/0001688	12/2003	Shen	N/A	N/A
2004/0023674	12/2003	Miller	N/A	N/A
2004/0127214	12/2003	Reddy et al.	N/A	N/A
2004/0137915	12/2003	Diener et al.	N/A	N/A
2004/0147254	12/2003	Reddy et al.	N/A	N/A
2004/0171390	12/2003	Chitrapu	N/A	N/A
2004/0203725	12/2003	Lahav et al.	N/A	N/A

2004/0203826	12/2003	Sugar et al.	N/A	N/A
2004/0208238	12/2003	Thomas et al.	N/A	N/A
2004/0219885	12/2003	Sugar et al.	N/A	N/A
2004/0233100	12/2003	Dibble et al.	N/A	N/A
2005/0003828	12/2004	Sugar et al.	N/A	N/A
2005/0096026	12/2004	Chitrapu et al.	N/A	N/A
2005/0107102	12/2004	Yoon et al.	N/A	N/A
2005/0114023	12/2004	Williamson et al.	N/A	N/A
2005/0152317	12/2004	Awater et al.	N/A	N/A
2005/0159928	12/2004	Moser	N/A	N/A
2005/0176401	12/2004	Nanda et al.	N/A	N/A
2005/0185618	12/2004	Friday et al.	N/A	N/A
2005/0192727	12/2004	Shostak et al.	N/A	N/A
2005/0227625	12/2004	Diener	N/A	N/A
2005/0285792	12/2004	Sugar et al.	N/A	N/A
2006/0025118	12/2005	Chitrapu et al.	N/A	N/A
2006/0047704	12/2005	Gopalakrishnan	N/A	N/A
2006/0074558	12/2005	Williamson et al.	N/A	N/A
2006/0080040	12/2005	Garczarek et al.	N/A	N/A
2006/0111899	12/2005	Padhi et al.	N/A	N/A
2006/0128311	12/2005	Tesfai	N/A	N/A
2006/0133263	12/2005	Bernard et al.	N/A	N/A
2006/0199546	12/2005	Durgin	N/A	N/A
2006/0235574	12/2005	Lapinski et al.	N/A	N/A
2006/0238417	12/2005	Jendbro et al.	N/A	N/A
2006/0258347	12/2005	Chitrapu	N/A	N/A
2007/0016412	12/2006	Mehrotra et al.	N/A	N/A
2007/0041481	12/2006	Malkemes et al.	N/A	N/A
2007/0049823	12/2006	Li	N/A	N/A
2007/0076657	12/2006	Woodings et al.	N/A	N/A
2007/0098089	12/2006	Li et al.	N/A	N/A
2007/0111746	12/2006	Anderson	N/A	N/A
2007/0149216	12/2006	Misikangas	N/A	N/A
2007/0168580	12/2006	Schumacher	N/A	N/A
2007/0171889	12/2006	Kwon et al.	N/A	N/A
2007/0203645	12/2006	Dees et al.	N/A	N/A
2007/0223419	12/2006	Ji et al.	N/A	N/A
2007/0233336	12/2006	Serguei et al.	N/A	N/A
2007/0233409	12/2006	Boyan et al.	N/A	N/A
2007/0273581	12/2006	Garrison et al.	N/A	N/A
2007/0293171	12/2006	Li et al.	N/A	N/A
2007/0296591	12/2006	Frederick et al.	N/A	N/A
2007/0297541	12/2006	Mcgehee	N/A	N/A
2008/0001735	12/2007	Tran	N/A	N/A
2008/0010040	12/2007	Mcgehee	N/A	N/A
2008/0090563	12/2007	Chitrapu	N/A	N/A
2008/0113634	12/2007	Gates et al.	N/A	N/A
2008/0123731	12/2007	Wegener	N/A	N/A
2008/0129367	12/2007	Murata et al.	N/A	N/A
2008/0130519	12/2007	Bahl et al.	N/A	N/A

2008/0133190	12/2007	Peretz et al.	N/A	N/A
2008/0180325	12/2007	Chung et al.	N/A	N/A
2008/0186235	12/2007	Struckman et al.	N/A	N/A
2008/0195584	12/2007	Nath et al.	N/A	N/A
2008/0209117	12/2007	Kajigaya	N/A	N/A
2008/0211481	12/2007	Chen	N/A	N/A
2008/0214903	12/2007	Orbach	N/A	N/A
2008/0252516	12/2007	Ho et al.	N/A	N/A
2008/0261509	12/2007	Sen	N/A	N/A
2008/0293353	12/2007	Mody et al.	N/A	N/A
2009/0006103	12/2008	Koishida et al.	N/A	N/A
2009/0011713	12/2008	Abusubaih et al.	N/A	N/A
2009/0018422	12/2008	Banet et al.	N/A	N/A
2009/0021420	12/2008	Sahinoglu	N/A	N/A
2009/0046003	12/2008	Tung et al.	N/A	N/A
2009/0046625	12/2008	Diener et al.	N/A	N/A
2009/0066578	12/2008	Beadle et al.	N/A	N/A
2009/0086993	12/2008	Kawaguchi et al.	N/A	N/A
2009/0103094	12/2008	Hilfiker et al.	N/A	N/A
2009/0111463	12/2008	Simms et al.	N/A	N/A
2009/0131067	12/2008	Aaron	N/A	N/A
2009/0135046	12/2008	Steele et al.	N/A	N/A
2009/0136052	12/2008	Hohlfeld et al.	N/A	N/A
2009/0143019	12/2008	Shellhammer	N/A	N/A
2009/0146881	12/2008	Mesecher	N/A	N/A
2009/0149202	12/2008	Hill et al.	N/A	N/A
2009/0190511	12/2008	Li et al.	N/A	N/A
2009/0207950	12/2008	Tsuruta et al.	N/A	N/A
2009/0224957	12/2008	Chung et al.	N/A	N/A
2009/0245327	12/2008	Michaels	N/A	N/A
2009/0278733	12/2008	Haworth	N/A	N/A
2009/0280748	12/2008	Shan et al.	N/A	N/A
2009/0282130	12/2008	Antoniou et al.	N/A	N/A
2009/0285173	12/2008	Koorapaty et al.	N/A	N/A
2009/0286563	12/2008	Ji et al.	N/A	N/A
2009/0322510	12/2008	Berger et al.	N/A	N/A
2010/0020707	12/2009	Woodings	N/A	N/A
2010/0044122	12/2009	Sleeman et al.	N/A	N/A
2010/0056200	12/2009	Tolonen	N/A	N/A
2010/0075704	12/2009	Mchenry et al.	N/A	N/A
2010/0109936	12/2009	Levy	N/A	N/A
2010/0142454	12/2009	Chang	N/A	N/A
2010/0150122	12/2009	Berger et al.	N/A	N/A
2010/0172443	12/2009	Shim et al.	N/A	N/A
2010/0173586	12/2009	Mchenry et al.	N/A	N/A
2010/0176988	12/2009	Maezawa et al.	N/A	N/A
2010/0177710	12/2009	Gutkin et al.	N/A	N/A
2010/0220011	12/2009	Heuser	N/A	N/A
2010/0253512	12/2009	Wagner et al.	N/A	N/A
2010/0255794	12/2009	Agnew	N/A	N/A

2010/0255801	12/2009	Gunasekara et al.	N/A	N/A
2010/0259998	12/2009	Kwon et al.	N/A	N/A
2010/0279680	12/2009	Reudink	N/A	N/A
2010/0292930	12/2009	Koster et al.	N/A	N/A
2010/0306249	12/2009	Hill et al.	N/A	N/A
2010/0309317	12/2009	Wu et al.	N/A	N/A
2010/0325621	12/2009	Andrade et al.	N/A	N/A
2011/0022342	12/2010	Pandharipande et al.	N/A	N/A
2011/0045781	12/2010	Shellhammer et al.	N/A	N/A
2011/0053604	12/2010	Kim et al.	N/A	N/A
2011/0059747	12/2010	Lindoff et al.	N/A	N/A
2011/0070885	12/2010	Ruuska et al.	N/A	N/A
2011/0074631	12/2010	Parker	N/A	N/A
2011/0077017	12/2010	Yu et al.	N/A	N/A
2011/0087639	12/2010	Gurney	N/A	N/A
2011/0090939	12/2010	Diener et al.	N/A	N/A
2011/0096770	12/2010	Henry	N/A	N/A
2011/0102258	12/2010	Underbrink et al.	N/A	N/A
2011/0111751	12/2010	Markhovsky et al.	N/A	N/A
2011/0116484	12/2010	Henry	N/A	N/A
2011/0117869	12/2010	Woodings	N/A	N/A
2011/0122855	12/2010	Henry	N/A	N/A
2011/0129006	12/2010	Jung et al.	N/A	N/A
2011/0131260	12/2010	Mody	N/A	N/A
2011/0134878	12/2010	Geiger et al.	N/A	N/A
2011/0136478	12/2010	Trigui	N/A	N/A
2011/0151876	12/2010	Ishii et al.	N/A	N/A
2011/0183621	12/2010	Quan et al.	N/A	N/A
2011/0183685	12/2010	Burton et al.	N/A	N/A
2011/0185059	12/2010	Adnani et al.	N/A	N/A
2011/0188544	12/2010	Ponnuswamy	N/A	N/A
2011/0235728	12/2010	Karabinis	N/A	N/A
2011/0237243	12/2010	Guvenc et al.	N/A	N/A
2011/0241923	12/2010	Chernukhin	N/A	N/A
2011/0273328	12/2010	Parker	N/A	N/A
2011/0286555	12/2010	Cho et al.	N/A	N/A
2011/0286604	12/2010	Matsuo	N/A	N/A
2011/0287779	12/2010	Harper	N/A	N/A
2011/0299481	12/2010	Kim et al.	N/A	N/A
2011/0300849	12/2010	Chan	N/A	N/A
2011/0319120	12/2010	Chen et al.	N/A	N/A
2012/0014332	12/2011	Smith et al.	N/A	N/A
2012/0032854	12/2011	Bull et al.	N/A	N/A
2012/0039284	12/2011	Barbieri et al.	N/A	N/A
2012/0040602	12/2011	Charland	N/A	N/A
2012/0047544	12/2011	Bouchard	N/A	N/A
2012/0052869	12/2011	Lindoff et al.	N/A	N/A
2012/0058775	12/2011	Dupray et al.	N/A	N/A
2012/0063302	12/2011	Damnjanovic et al.	N/A	N/A
2012/0071188	12/2011	Wang et al.	N/A	N/A

2012/0072986	12/2011	Livsics et al.	N/A	N/A
2012/0077510	12/2011	Chen et al.	N/A	N/A
2012/0078071	12/2011	Bohm et al.	N/A	N/A
2012/0081248	12/2011	Kennedy et al.	N/A	N/A
2012/0094681	12/2011	Freda et al.	N/A	N/A
2012/0100810	12/2011	Oksanen et al.	N/A	N/A
2012/0105066	12/2011	Marvin et al.	N/A	N/A
2012/0115522	12/2011	Nama et al.	N/A	N/A
2012/0115525	12/2011	Kang et al.	N/A	N/A
2012/0120892	12/2011	Freda et al.	N/A	N/A
2012/0129522	12/2011	Kim et al.	N/A	N/A
2012/0140236	12/2011	Babbitt et al.	N/A	N/A
2012/0142386	12/2011	Mody et al.	N/A	N/A
2012/0148068	12/2011	Chandra et al.	N/A	N/A
2012/0148069	12/2011	Bai et al.	N/A	N/A
2012/0155217	12/2011	Dellinger et al.	N/A	N/A
2012/0169424	12/2011	Pinarello et al.	N/A	N/A
2012/0182430	12/2011	Birkett et al.	N/A	N/A
2012/0195269	12/2011	Kang et al.	N/A	N/A
2012/0212628	12/2011	Wu et al.	N/A	N/A
2012/0214511	12/2011	Vartanian et al.	N/A	N/A
2012/0230214	12/2011	Kozisek et al.	N/A	N/A
2012/0246392	12/2011	Cheon	N/A	N/A
2012/0264388	12/2011	Guo et al.	N/A	N/A
2012/0264445	12/2011	Lee et al.	N/A	N/A
2012/0275354	12/2011	Villain	N/A	N/A
2012/0281000	12/2011	Woodings	N/A	N/A
2012/0282942	12/2011	Uusitalo et al.	N/A	N/A
2012/0295575	12/2011	Nam	N/A	N/A
2012/0302190	12/2011	Mchenry	N/A	N/A
2012/0302263	12/2011	Tinnakornsrisuphap et al.	N/A	N/A
2012/0309288	12/2011	Lu	N/A	N/A
2012/0321024	12/2011	Wasiewicz et al.	N/A	N/A
2012/0322487	12/2011	Stanforth	N/A	N/A
2013/0005240	12/2012	Novak et al.	N/A	N/A
2013/0005374	12/2012	Uusitalo et al.	N/A	N/A
2013/0012134	12/2012	Jin et al.	N/A	N/A
2013/0017794	12/2012	Kloper et al.	N/A	N/A
2013/0023285	12/2012	Markhovsky et al.	N/A	N/A
2013/0028111	12/2012	Dain et al.	N/A	N/A
2013/0035108	12/2012	Joslyn et al.	N/A	N/A
2013/0035128	12/2012	Chan et al.	N/A	N/A
2013/0045754	12/2012	Markhovsky et al.	N/A	N/A
2013/0052939	12/2012	Anniballi et al.	N/A	N/A
2013/0053054	12/2012	Lovitt et al.	N/A	N/A
2013/0062334	12/2012	Bilchinsky et al.	N/A	N/A
2013/0064197	12/2012	Novak et al.	N/A	N/A
2013/0064328	12/2012	Adnani et al.	N/A	N/A
2013/0070639	12/2012	Demura et al.	N/A	N/A

2013/0090071	12/2012	Abraham et al.	N/A	N/A
2013/0095843	12/2012	Smith et al.	N/A	N/A
2013/0100154	12/2012	Woodings et al.	N/A	N/A
2013/0103684	12/2012	Yee et al.	N/A	N/A
2013/0113659	12/2012	Morgan	N/A	N/A
2013/0165051	12/2012	Li et al.	N/A	N/A
2013/0165134	12/2012	Touag et al.	N/A	N/A
2013/0165170	12/2012	Kang	N/A	N/A
2013/0183989	12/2012	Hasegawa et al.	N/A	N/A
2013/0183994	12/2012	Ringstroem et al.	N/A	N/A
2013/0184022	12/2012	Schmidt	N/A	N/A
2013/0190003	12/2012	Smith et al.	N/A	N/A
2013/0190028	12/2012	Wang et al.	N/A	N/A
2013/0196677	12/2012	Smith et al.	N/A	N/A
2013/0208587	12/2012	Bala et al.	N/A	N/A
2013/0210457	12/2012	Kummetz	N/A	N/A
2013/0210473	12/2012	Weigand	N/A	N/A
2013/0217406	12/2012	Villard et al.	N/A	N/A
2013/0217408	12/2012	Difazio et al.	N/A	N/A
2013/0217450	12/2012	Kanj et al.	N/A	N/A
2013/0231121	12/2012	Kwak et al.	N/A	N/A
2013/0237212	12/2012	Khayrallah et al.	N/A	N/A
2013/0242792	12/2012	Woodings	N/A	N/A
2013/0242934	12/2012	Ueda et al.	N/A	N/A
2013/0260703	12/2012	Actis et al.	N/A	N/A
2013/0265198	12/2012	Stroud	N/A	N/A
2013/0272436	12/2012	Makhlouf et al.	N/A	N/A
2013/0275346	12/2012	Srikanteswara	706/12	H04W 74/002
2013/0279556	12/2012	Seller	N/A	N/A
2013/0288734	12/2012	Mody et al.	N/A	N/A
2013/0309975	12/2012	Kpodzo et al.	N/A	N/A
2013/0315112	12/2012	Gormley et al.	N/A	N/A
2013/0329690	12/2012	Kim et al.	N/A	N/A
2013/0331114	12/2012	Gormley et al.	N/A	N/A
2014/0003547	12/2013	Williams et al.	N/A	N/A
2014/0015796	12/2013	Philipp	N/A	N/A
2014/0018683	12/2013	Park et al.	N/A	N/A
2014/0024405	12/2013	Qiu	N/A	N/A
2014/0064723	12/2013	Adles et al.	N/A	N/A
2014/0066060	12/2013	Ngai	N/A	N/A
2014/0073261	12/2013	Hassan et al.	N/A	N/A
2014/0086212	12/2013	Kafle et al.	N/A	N/A
2014/0128032	12/2013	Muthukumar et al.	N/A	N/A
2014/0139374	12/2013	Wellman et al.	N/A	N/A
2014/0163309	12/2013	Bernhard et al.	N/A	N/A
2014/0199993	12/2013	Dhanda et al.	N/A	N/A
2014/0201367	12/2013	Trummer et al.	N/A	N/A
2014/0204766	12/2013	Immendorf et al.	N/A	N/A
2014/0206279	12/2013	Immendorf et al.	N/A	N/A

2014/0206307	12/2013	Maurer et al.	N/A	N/A
2014/0206343	12/2013	Immendorf et al.	N/A	N/A
2014/0207414	12/2013	Bordow	N/A	N/A
2014/0225590	12/2013	Jacobs	N/A	N/A
2014/0256268	12/2013	Olgaard	N/A	N/A
2014/0256370	12/2013	Gautier et al.	N/A	N/A
2014/0269374	12/2013	Abdelmonem et al.	N/A	N/A
2014/0269376	12/2013	Garcia et al.	N/A	N/A
2014/0274103	12/2013	Steer et al.	N/A	N/A
2014/0287100	12/2013	Libman	N/A	N/A
2014/0301216	12/2013	Immendorf et al.	N/A	N/A
2014/0302796	12/2013	Gormley et al.	N/A	N/A
2014/0335879	12/2013	Immendorf et al.	N/A	N/A
2014/0340684	12/2013	Edler et al.	N/A	N/A
2014/0342675	12/2013	Massarella et al.	N/A	N/A
2014/0348004	12/2013	Ponnuswamy	N/A	N/A
2014/0362934	12/2013	Kumar	N/A	N/A
2015/0016429	12/2014	Menon et al.	N/A	N/A
2015/0023329	12/2014	Jiang et al.	N/A	N/A
2015/0068296	12/2014	Lanza di Scalea	N/A	N/A
2015/0072633	12/2014	Massarella et al.	N/A	N/A
2015/0126181	12/2014	Breuer et al.	N/A	N/A
2015/0133058	12/2014	Livis et al.	N/A	N/A
2015/0150753	12/2014	Racette	N/A	N/A
2015/0156827	12/2014	Ibragimov et al.	N/A	N/A
2015/0170145	12/2014	Patel et al.	N/A	N/A
2015/0201385	12/2014	Mercer et al.	N/A	N/A
2015/0208312	12/2014	Fudaba	N/A	N/A
2015/0215794	12/2014	Gormley et al.	N/A	N/A
2015/0215949	12/2014	Gormley et al.	N/A	N/A
2015/0242384	12/2014	Reiter	N/A	N/A
2015/0248047	12/2014	Chakraborty	N/A	N/A
2015/0289254	12/2014	Garcia et al.	N/A	N/A
2015/0289265	12/2014	Gormley et al.	N/A	N/A
2015/0296386	12/2014	Menon et al.	N/A	N/A
2015/0301167	12/2014	Sentelle et al.	N/A	N/A
2015/0319768	12/2014	Abdelmonem et al.	N/A	N/A
2015/0373528	12/2014	Iwai	455/419	H04W 4/70
2016/0014713	12/2015	Kennedy et al.	N/A	N/A
2016/0037550	12/2015	Barabell et al.	N/A	N/A
2016/0050690	12/2015	Yun et al.	N/A	N/A
2016/0061583	12/2015	Ryu et al.	N/A	N/A
2016/0069994	12/2015	Allen et al.	N/A	N/A
2016/0073318	12/2015	Aguirre	N/A	N/A
2016/0086621	12/2015	Hearing et al.	N/A	N/A
2016/0095188	12/2015	Verberkt et al.	N/A	N/A
2016/0117853	12/2015	Zhong et al.	N/A	N/A
2016/0124071	12/2015	Baxley et al.	N/A	N/A
2016/0126988	12/2015	Mester et al.	N/A	N/A

2016/0127110	12/2015	McMeekin et al.	N/A	N/A
2016/0127392	12/2015	Baxley et al.	N/A	N/A
2016/0154406	12/2015	Im et al.	N/A	N/A
2016/0161578	12/2015	Weissler et al.	N/A	N/A
2016/0198471	12/2015	Young et al.	N/A	N/A
2016/0219506	12/2015	Pratt et al.	N/A	N/A
2016/0219590	12/2015	Khawer et al.	N/A	N/A
2016/0225240	12/2015	Voddhi et al.	N/A	N/A
2016/0245921	12/2015	Li et al.	N/A	N/A
2016/0252342	12/2015	Feldmann et al.	N/A	N/A
2016/0334527	12/2015	Xu et al.	N/A	N/A
2016/0345135	12/2015	Garcia et al.	N/A	N/A
2016/0364079	12/2015	Qiu et al.	N/A	N/A
2016/0366685	12/2015	Gormley et al.	N/A	N/A
2016/0374088	12/2015	Garcia et al.	N/A	N/A
2017/0024767	12/2016	Johnson, Jr. et al.	N/A	N/A
2017/0025996	12/2016	Cheung et al.	N/A	N/A
2017/0039413	12/2016	Nadler	N/A	N/A
2017/0048838	12/2016	Chrisikos et al.	N/A	N/A
2017/0061690	12/2016	Laughlin et al.	N/A	N/A
2017/0064564	12/2016	Yun et al.	N/A	N/A
2017/0078792	12/2016	Simons	N/A	N/A
2017/0079007	12/2016	Carbajal	N/A	N/A
2017/0094527	12/2016	Shattil et al.	N/A	N/A
2017/0118049	12/2016	Miao et al.	N/A	N/A
2017/0134631	12/2016	Zhao et al.	N/A	N/A
2017/0146462	12/2016	Baker et al.	N/A	N/A
2017/0148332	12/2016	Ziemba et al.	N/A	N/A
2017/0148467	12/2016	Franklin et al.	N/A	N/A
2017/0192089	12/2016	Parker et al.	N/A	N/A
2017/0234979	12/2016	Mathews et al.	N/A	N/A
2017/0237484	12/2016	Heath et al.	N/A	N/A
2017/0238201	12/2016	Gormley et al.	N/A	N/A
2017/0238203	12/2016	Dzierwa et al.	N/A	N/A
2017/0243138	12/2016	Dzierwa et al.	N/A	N/A
2017/0243139	12/2016	Dzierwa et al.	N/A	N/A
2017/0248677	12/2016	Mahmood et al.	N/A	N/A
2017/0248807	12/2016	Jiang et al.	N/A	N/A
2017/0250766	12/2016	Dzierwa et al.	N/A	N/A
2017/0253330	12/2016	Saigh et al.	N/A	N/A
2017/0261604	12/2016	Van Voorst	N/A	N/A
2017/0261613	12/2016	Van Voorst	N/A	N/A
2017/0261615	12/2016	Mng et al.	N/A	N/A
2017/0274992	12/2016	Chretien	N/A	N/A
2017/0289840	12/2016	Sung et al.	N/A	N/A
2017/0290075	12/2016	Carbajal et al.	N/A	N/A
2017/0311307	12/2016	Negus et al.	N/A	N/A
2017/0358103	12/2016	Shao et al.	N/A	N/A
2017/0366361	12/2016	Afkhami et al.	N/A	N/A
2017/0374572	12/2016	Kleinbeck et al.	N/A	N/A

2017/0374573	12/2016	Kleinbeck et al.	N/A	N/A
2018/0006730	12/2017	Kuo et al.	N/A	N/A
2018/0014217	12/2017	Kleinbeck et al.	N/A	N/A
2018/0024220	12/2017	Massarella et al.	N/A	N/A
2018/0070362	12/2017	Ryan et al.	N/A	N/A
2018/0074170	12/2017	Ray	N/A	N/A
2018/0074171	12/2017	Ray et al.	N/A	N/A
2018/0081355	12/2017	Magy et al.	N/A	N/A
2018/0083721	12/2017	Wada et al.	N/A	N/A
2018/0124533	12/2017	Messier et al.	N/A	N/A
2018/0129881	12/2017	Seeber et al.	N/A	N/A
2018/0131445	12/2017	Esman et al.	N/A	N/A
2018/0143314	12/2017	Pelletier et al.	N/A	N/A
2018/0149729	12/2017	Grandin et al.	N/A	N/A
2018/0211179	12/2017	Dzierwa	N/A	N/A
2018/0284758	12/2017	Cella et al.	N/A	N/A
2018/0288620	12/2017	Jayawickrama et al.	N/A	N/A
2018/0294901	12/2017	Garcia et al.	N/A	N/A
2018/0313877	12/2017	Brant et al.	N/A	N/A
2018/0313945	12/2017	Parker et al.	N/A	N/A
2018/0324595	12/2017	Shima	N/A	N/A
2018/0329020	12/2017	Hafizovic et al.	N/A	N/A
2018/0331863	12/2017	Carbajal	N/A	N/A
2019/0004518	12/2018	Zhou et al.	N/A	N/A
2019/0018103	12/2018	Qian et al.	N/A	N/A
2019/0064130	12/2018	Kanazawa et al.	N/A	N/A
2019/0064223	12/2018	Kincaid	N/A	N/A
2019/0072601	12/2018	Dzierwa et al.	N/A	N/A
2019/0074802	12/2018	Geha et al.	N/A	N/A
2019/0077507	12/2018	Ferris et al.	N/A	N/A
2019/0123428	12/2018	Packer et al.	N/A	N/A
2019/0180630	12/2018	Kleinbeck	N/A	N/A
2019/0191313	12/2018	Dzierwa et al.	N/A	N/A
2019/0200303	12/2018	Nakahara	N/A	N/A
2019/0208112	12/2018	Kleinbeck	N/A	N/A
2019/0208491	12/2018	Dzierwa et al.	N/A	N/A
2019/0215709	12/2018	Kleinbeck et al.	N/A	N/A
2019/0223139	12/2018	Kleinbeck et al.	N/A	N/A
2019/0230539	12/2018	Dzierwa et al.	N/A	N/A
2019/0230540	12/2018	Carbajal et al.	N/A	N/A
2019/0236266	12/2018	Nashimoto et al.	N/A	N/A
2019/0245722	12/2018	Carbajal	N/A	N/A
2019/0246304	12/2018	Dzierwa et al.	N/A	N/A
2019/0253160	12/2018	Garcia et al.	N/A	N/A
2019/0253905	12/2018	Kleinbeck et al.	N/A	N/A
2019/0260768	12/2018	Mestha et al.	N/A	N/A
2019/0274059	12/2018	Kleinbeck et al.	N/A	N/A
2019/0296910	12/2018	Cheung	N/A	N/A
2019/0302249	12/2018	High et al.	N/A	N/A
2019/0342202	12/2018	Ryan et al.	N/A	N/A

2019/0346571	12/2018	Furumoto	N/A	N/A
2019/0360783	12/2018	Whittaker	N/A	N/A
2019/0364433	12/2018	Das	N/A	G06N 5/045
2019/0364533	12/2018	Kleinbeck et al.	N/A	N/A
2020/0034620	12/2019	Lutterodt	N/A	N/A
2020/0036459	12/2019	Menon	N/A	N/A
2020/0036487	12/2019	Hammond et al.	N/A	N/A
2020/0043346	12/2019	Vacek	N/A	N/A
2020/0059800	12/2019	Menon et al.	N/A	N/A
2020/0066132	12/2019	Kleinbeck	N/A	N/A
2020/0067752	12/2019	DelMarco	N/A	N/A
2020/0068573	12/2019	Drozdz et al.	N/A	N/A
2020/0096548	12/2019	Dzierwa et al.	N/A	N/A
2020/0107207	12/2019	Kleinbeck et al.	N/A	N/A
2020/0120266	12/2019	Kleinbeck	N/A	N/A
2020/0128418	12/2019	Dzierwa et al.	N/A	N/A
2020/0137583	12/2019	Economy et al.	N/A	N/A
2020/0142029	12/2019	Brooker et al.	N/A	N/A
2020/0145032	12/2019	Ayala et al.	N/A	N/A
2020/0162890	12/2019	Spencer et al.	N/A	N/A
2020/0167196	12/2019	Smith et al.	N/A	N/A
2020/0169892	12/2019	Dzierwa et al.	N/A	N/A
2020/0184832	12/2019	Kleinbeck	N/A	N/A
2020/0196269	12/2019	Dzierwa et al.	N/A	N/A
2020/0196270	12/2019	Kleinbeck et al.	N/A	N/A
2020/0242603	12/2019	Salkintzis	N/A	N/A
2020/0245167	12/2019	Kleinbeck et al.	N/A	N/A
2020/0260306	12/2019	Kleinbeck et al.	N/A	N/A
2020/0294032	12/2019	Cheng et al.	N/A	N/A
2020/0295855	12/2019	Kleinbeck et al.	N/A	N/A
2020/0382961	12/2019	Shattil et al.	N/A	N/A
2020/0388036	12/2019	Skrede et al.	N/A	N/A
2021/0014696	12/2020	Brookes	N/A	N/A
2021/0067974	12/2020	Guo et al.	N/A	N/A
2021/0082254	12/2020	Givant	N/A	N/A
2021/0084217	12/2020	Kleinbeck	N/A	N/A
2021/0211911	12/2020	Kleinbeck et al.	N/A	N/A
2021/0250795	12/2020	Dzierwa et al.	N/A	N/A
2021/0255356	12/2020	Vu et al.	N/A	N/A
2021/0280039	12/2020	Kleinbeck	N/A	N/A
2021/0281510	12/2020	Brown, Jr. et al.	N/A	N/A
2021/0306022	12/2020	Fernando et al.	N/A	N/A
2021/0360423	12/2020	Dzierwa et al.	N/A	N/A
2021/0360450	12/2020	Kleinbeck et al.	N/A	N/A
2021/0360453	12/2020	Kleinbeck et al.	N/A	N/A
2021/0360454	12/2020	Carbajal et al.	N/A	N/A
2021/0409591	12/2020	Kleinbeck	N/A	N/A
2022/0030541	12/2021	Dzierwa et al.	N/A	N/A
2022/0052770	12/2021	Kleinbeck et al.	N/A	N/A

2022/0128612	12/2021	Dzierwa et al.	N/A	N/A
2022/0131623	12/2021	Garcia et al.	N/A	N/A
2022/0150824	12/2021	Kleinbeck et al.	N/A	N/A
2022/0174525	12/2021	Dzierwa et al.	N/A	N/A
2022/0253407	12/2021	Tran	N/A	N/A
2022/0262228	12/2021	Kleinbeck	N/A	N/A
2022/0262261	12/2021	Kleinbeck	N/A	N/A
2022/0286997	12/2021	Kleinbeck et al.	N/A	N/A
2022/0376921	12/2021	Maria	N/A	N/A
2023/0018133	12/2022	Burnette et al.	N/A	N/A
2023/0087729	12/2022	Goldstein et al.	N/A	N/A
2023/0098387	12/2022	Hafeez et al.	N/A	N/A
2023/0105718	12/2022	Carbajal	N/A	N/A
2023/0114804	12/2022	Kleinbeck	N/A	N/A
2023/0116761	12/2022	Barry et al.	N/A	N/A
2023/0118723	12/2022	Carbajal et al.	N/A	N/A
2023/0123375	12/2022	Dzierwa et al.	N/A	N/A
2023/0126223	12/2022	Kleinbeck et al.	N/A	N/A
2023/0189010	12/2022	Muhammad et al.	N/A	N/A
2023/0189382	12/2022	Haustein	370/329	H04W 76/20
2023/0206216	12/2022	Lehmann et al.	N/A	N/A
2023/0209378	12/2022	Kleinbeck et al.	N/A	N/A
2023/0232244	12/2022	Dzierwa et al.	N/A	N/A
2023/0252744	12/2022	Miller et al.	N/A	N/A
2023/0254054	12/2022	Garcia et al.	N/A	N/A
2023/0254567	12/2022	Kleinbeck	N/A	N/A
2023/0254702	12/2022	Damnjanovic et al.	N/A	N/A
2023/0275791	12/2022	Carbajal	N/A	N/A
2023/0276280	12/2022	Kleinbeck et al.	N/A	N/A
2023/0300256	12/2022	Zhao et al.	N/A	N/A
2023/0308789	12/2022	Tian et al.	N/A	N/A
2023/0308915	12/2022	Carbajal et al.	N/A	N/A
2023/0326323	12/2022	Kleinbeck	N/A	N/A
2023/0334472	12/2022	Pene	N/A	N/A
2023/0345441	12/2022	Baxley	N/A	N/A
2023/0349962	12/2022	Dzierwa et al.	N/A	N/A
2023/0378645	12/2022	Tran	N/A	N/A
2023/0403564	12/2022	Dzierwa et al.	N/A	N/A
2024/0007204	12/2023	Kleinbeck et al.	N/A	N/A
2024/0023054	12/2023	Dzierwa et al.	N/A	N/A
2024/0029572	12/2023	Kleinbeck	N/A	N/A
2024/0031042	12/2023	Garcia et al.	N/A	N/A
2024/0032084	12/2023	Hellwig	N/A	N/A
2024/0089130	12/2023	Wang et al.	N/A	N/A
2024/0097951	12/2023	Carbajal	N/A	N/A
2024/0103059	12/2023	Dzierwa et al.	N/A	N/A
2024/0114370	12/2023	Kleinbeck et al.	N/A	N/A
2024/0160791	12/2023	He et al.	N/A	N/A
2024/0242589	12/2023	Kleinbeck	N/A	N/A

2024/0267268	12/2023	Carbajal	N/A	N/A
2024/0267769	12/2023	Dzierwa et al.	N/A	N/A
2024/0275504	12/2023	Kleinbeck et al.	N/A	N/A
2024/0276261	12/2023	Kleinbeck et al.	N/A	N/A
2024/0284214	12/2023	Kleinbeck et al.	N/A	N/A
2024/0292248	12/2023	Kleinbeck et al.	N/A	N/A
2024/0330927	12/2023	Abdelrahman et al.	N/A	N/A
2024/0362631	12/2023	Agrawal et al.	N/A	N/A
2024/0373463	12/2023	Furuichi et al.	N/A	N/A
2024/0378982	12/2023	Kleinbeck	N/A	N/A
2024/0381106	12/2023	Montalvo et al.	N/A	N/A
2024/0386800	12/2023	Kleinbeck et al.	N/A	N/A
2024/0386801	12/2023	Kleinbeck et al.	N/A	N/A
2024/0388937	12/2023	Murias et al.	N/A	N/A
2024/0430142	12/2023	Carbajal	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110035440	12/2018	CN	N/A
115482071	12/2021	CN	N/A
117196617	12/2022	CN	N/A
117998368	12/2023	CN	N/A
100248671	12/1999	KR	N/A
20140041618	12/2013	KR	N/A
953557	12/1981	SU	N/A
2012129932	12/2011	WO	N/A
2018184682	12/2017	WO	N/A
2023197982	12/2022	WO	N/A
2023232010	12/2022	WO	N/A

OTHER PUBLICATIONS

RF and Digital Signal Processing for Software-Defined Radio, Chapter 4—High-Level Requirements and Link Budget Analysis (Year: 2009). cited by applicant

“A Hardware Design for Time Delay Estimation of TDOA”; Li et al.; 2013 IEEE International Conference on Signal Processing, Communication and Computing (ICSPCC 2013); Aug. 2013 (Year: 2013). cited by applicant

“A Low-Cost, Near-Real-Time Two-LIAS-Based UWB Emitter Monitoring System”; Wang et al.; IEEE A&E Systems Magazine Nov. 2015 (Year: 2015). cited by applicant

“Joint TDOA and FDOA Estimation: A Conditional Bound and Its Use for Optimally Weighted Localization”; Yeredor et al.; IEEE Transactions on Signal Processing, vol. 59, No. 4, Apr. 2011 (Year: 2011). cited by applicant

“Multipath TDOA and FDOA Estimation Using the EM Algorithm”; Belanger; Apr. 27, 1993; 1993 IEEE International Conference on Acoustics, Speech, and Signal Processing (Year: 1993). cited by applicant

“Noise Figure”, Wikipedia, located at https://en.wikipedia.org/wiki/Noise_figure (Year: 2022). cited by applicant

“Signal Models for TDOA/FDOA Estimation”; Fowler et al.; IEEE Transactions on Aerospace and Electronic Systems vol. 44, No. Oct. 4, 2008 (Year: 2008). cited by applicant

“Specific attenuation model for rain for use in prediction methods”, Recommendation ITU-R p.

838-3 (Year: 2005). cited by applicant

Bluetooth vs Zigbee-difference between Bluetooth and Zigbee (located at <https://www.rfwireless-world.com/Terminology/Bluetooth-vs-zigbee.html>) (Year: 2012). cited by applicant

Boll S.F., Suppression of Acoustic Noise in Speech Using Spectral Subtraction, Apr. 1979, IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. ASSP-27, No. 2, (Year: 1979). cited by applicant

David Eppink and Wolf Kuebler, "TIREM/SEM Handbook", Mar. 1994, IIT Research Institute, p. 1-6, located at [http://www.dtic.mil/cgi-bin/GetTRDoc?](http://www.dtic.mil/cgi-bin/GetTRDoc?Location=U2&doc=GetTRDoc.pdf&AD=ADA296913)

[Location=U2&doc=GetTRDoc.pdf&AD=ADA296913](http://www.dtic.mil/cgi-bin/GetTRDoc?Location=U2&doc=GetTRDoc.pdf&AD=ADA296913). cited by applicant

Digital Global Systems, U.S. Appl. No. 18/784,394, filed Jul. 25, 2024, Non-Provisional Patent Application; Entire Document. cited by applicant

English translation of SU-953557-A1 (Year: 2024). cited by applicant

Gabriel Garcia and Daniel Carbajal, U.S. Appl. No. 61/789,758, Provisional Patent Application, filed Mar. 15, 2013 (Specification, Claims, and Drawings). cited by applicant

Gary L. Sugar, System and method for locating wireless devices in an unsynchronized wireless network, U.S. Appl. No. 60/319,737, Provisional Patent Application filed on Nov. 27, 2002, Specification including the claims, abstract, and drawings. cited by applicant

IEEE 100 the Authoritative Dictionary of IEEE Standards Terms. Seventh Edition. Published by Standards Information Network IEEE Press. p. 6 (Year: 2000). cited by applicant

International Search Report and Written Opinion dated Jun. 21, 2018 issued by the International Application Division, Korean Intellectual Property Office as International Searching Authority in connection with International Application No. PCT/US2018/014504 (21 pages). cited by applicant

Mehmet Ali Aygul, Ahmed Naeem, Huseyin Arslan. "Blind Signal Analysis—Wireless Communication Signals", located at <https://doi.org/10.1002/9781119764441.ch12> (Year: 2021). cited by applicant

Mobile Emitter Geolocation and Tracking Using TDOA and FDOA Measurements; Musicki et al.; IEEE Transactions on Signal Processing, vol. 58, No. 3, Mar. 2010 (Year: 2010). cited by applicant

S. Dörner, S. Cammerer, J. Hoydis and S. t. Brink, "Deep Learning Based Communication Over the Air," in IEEE Journal of Selected Topics in Signal Processing, vol. 12, No. 1, pp. 132-143, Feb. 2018, doi: 10.1109/JSTSP.2017.2784180. cited by applicant

Steven W. Smith, The Scientist & Engineer's Guide to Digital Signal Processing, 1999, California Technical Publishing, San Diego, California, 2nd Edition, p. 312 (located at http://www.analog.com/media/en/technical-documentation/dsp-book/dsp_book_ch18.pdf) (Year: 1999). cited by applicant

T. J. O'Shea, K. Karra and T. C. Clancy, "Learning to communicate: Channel auto-encoders, domain specific regularizers, and attention," 2016 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), Limassol, Cyprus, 2016, pp. 223-228, doi: 10.1109/ISSPIT.2016.7886039. cited by applicant

T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," in IEEE Transactions on Cognitive Communications and Networking, vol. 3, No. 4, pp. 563-575, Dec. 2017, doi: 10.1109/TCCN.2017.2758370. cited by applicant

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS (1) This application relates to and claims priority from the following U.S. patents and patent applications. This application is a continuation of U.S. application Ser. No. 18/825,699, filed Sep. 5, 2024, which is a continuation of U.S. application Ser. No. 18/638,122, filed Apr. 17, 2024, which is a continuation of U.S. application Ser. No. 18/525,040, filed Nov. 30, 2023, which is a continuation of U.S. application Ser. No. 18/142,917, filed May 3, 2023, which is a continuation of U.S. application Ser. No. 17/989,265, filed Nov. 17, 2022, which is a continuation of U.S. application Ser. No. 17/062,163, filed Oct. 2, 2020, which is a continuation of U.S. application Ser. No. 16/382,991 filed Apr. 12, 2019, which is a continuation of U.S. application Ser. No. 16/354,353 filed Mar. 15, 2019, which is a continuation of U.S. application Ser. No. 15/987,508 filed May 23, 2018, which is a continuation of U.S. application Ser. No. 15/492,475 filed Apr. 20, 2017, which is a continuation of U.S. application Ser. No. 14/504,819 filed Oct. 2, 2014, which is a continuation of U.S. application Ser. No. 14/329,838 filed Jul. 11, 2014, which is a continuation of U.S. application Ser. No. 14/273,157 filed May 8, 2014, which is a continuation of U.S. application Ser. No. 14/086,871 filed Nov. 21, 2013, which is a continuation-in-part of U.S. application Ser. No. 14/082,873 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/912,683 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety. U.S. application Ser. No. 14/086,871 is also a continuation-in-part of U.S. application Ser. No. 14/082,916 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/912,893 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety. U.S. application Ser. No. 14/086,871 is also a continuation-in-part of U.S. application Ser. No. 14/082,930 filed Nov. 18, 2013, which is a continuation of U.S. application Ser. No. 13/913,013 filed Jun. 7, 2013, which claims the benefit of U.S. Provisional Application No. 61/789,758 filed Mar. 15, 2013, each of which is hereby incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to spectrum analysis and management for radio frequency signals, and more particularly for automatically identifying open space in a wireless communications spectrum.

2. Description of the Prior Art

(2) Generally, it is known in the prior art to provide wireless communications spectrum management for detecting devices for managing the space. Spectrum management includes the process of regulating the use of radio frequencies to promote efficient use and gain net social benefit. A problem faced in effective spectrum management is the various numbers of devices emanating wireless signal propagations at different frequencies and across different technological standards. Coupled with the different regulations relating to spectrum usage around the globe effective spectrum management becomes difficult to obtain and at best can only be reached over a long period of time.

(3) Another problem facing effective spectrum management is the growing need from spectrum despite the finite amount of spectrum available. Wireless technologies have exponentially grown in recent years. Consequently, available spectrum has become a valuable resource that must be efficiently utilized. Therefore, systems and methods are needed to effectively manage and optimize the available spectrum that is being used.

(4) Most spectrum management devices may be categorized into two primary types. The first type is a spectral analyzer where a device is specifically fitted to run a 'scanner' type receiver that is tailored to provide spectral information for a narrow window of frequencies related to a specific and limited type of communications standard, such as cellular communication standard. Problems

arise with these narrowly tailored devices as cellular standards change and/or spectrum use changes impact the spectrum space of these technologies. Changes to the software and hardware for these narrowly tailored devices become too complicated, thus necessitating the need to purchase a totally different and new device. Unfortunately, this type of device is only for a specific use and cannot be used to alleviate the entire needs of the spectrum management community.

(5) The second type of spectral management device employs a methodology that requires bulky, extremely difficult to use processes, and expensive equipment. In order to attain a broad spectrum management view and complete all the necessary tasks, the device ends up becoming a conglomerate of software and hardware devices that is both hard to use and difficult to maneuver from one location to another.

(6) While there may be several additional problems associated with current spectrum management devices, at least four major problems exist overall: 1) most devices are built to inherently only handle specific spectrum technologies such as 900 MHz cellular spectrum while not being able to mitigate other technologies that may be interfering or competing with that spectrum, 2) the other spectrum management devices consist of large spectrum analyzers, database systems, and spectrum management software that is expensive, too bulky, and too difficult to manage for a user's basic needs, 3) other spectrum management devices in the prior art require external connectivity to remote databases to perform analysis and provide results or reports with analytics to aid in management of spectrum and/or devices, and 4) other devices of the prior art do not function to provide real-time or near real-time data and analysis to allow for efficient management of the space and/or devices and signals therein.

(7) Examples of relevant prior art documents include the following:

(8) U.S. Pat. No. RE43,066 for “System and method for reuse of communications spectrum for fixed and mobile applications with efficient method to mitigate interference” by inventor Mark Allen McHenry, filed Dec. 2, 2008, describes a communications system network enabling secondary use of spectrum on a non-interference basis. The system uses a modulation method to measure the background signals that eliminates self-generated interference and also identifies the secondary signal to all primary users via on/off amplitude modulation, allowing easy resolution of interference claims. The system uses high-processing gain probe waveforms that enable propagation measurements to be made with minimal interference to the primary users. The system measures background signals and identifies the types of nearby receivers and modifies the local frequency assignments to minimize interference caused by a secondary system due to non-linear mixing interference and interference caused by out-of-band transmitted signals (phase noise, harmonics, and spurs). The system infers a secondary node's elevation and mobility (thus, its probability to cause interference) by analysis of the amplitude of background signals. Elevated or mobile nodes are given more conservative frequency assignments than stationary nodes.

(9) U.S. Pat. No. 7,424,268 for “System and Method for Management of a Shared Frequency Band” by inventors Diener, et al., filed Apr. 22, 2003, discloses a system, method, software and related functions for managing activity in an unlicensed radio frequency band that is shared, both in frequency and time, by signals of multiple types. Signal pulse energy in the band is detected and is used to classify signals according to signal type. Using knowledge of the types of signals occurring in the frequency band and other spectrum activity related statistics (referred to as spectrum intelligence), actions can be taken in a device or network of devices to avoid interfering with other signals, and in general to optimize simultaneous use of the frequency band with the other signals. The spectrum intelligence may be used to suggest actions to a device user or network administrator, or to automatically invoke actions in a device or network of devices to maintain desirable performance.

(10) U.S. Pat. No. 8,249,631 for “Transmission power allocation/control method, communication device and program” by inventor Ryo Sawai, filed Jul. 21, 2010, teaches a method for allocating transmission power to a second communication service making secondary usage of a spectrum

assigned to a first communication service, in a node which is able to communicate with a secondary usage node. The method determines an interference power acceptable for two or more second communication services when the two or more second communication services are operated and allocates the transmission powers to the two or more second communication services.

(11) U.S. Pat. No. 8,094,610 for “Dynamic cellular cognitive system” by inventors Wang, et al., filed Feb. 25, 2009, discloses permitting high quality communications among a diverse set of cognitive radio nodes while minimizing interference to primary and other secondary users by employing dynamic spectrum access in a dynamic cellular cognitive system. Diverse device types interoperate, cooperate, and communicate with high spectrum efficiency and do not require infrastructure to form the network. The dynamic cellular cognitive system can expand to a wider geographical distribution via linking to existing infrastructure.

(12) U.S. Pat. No. 8,565,811 for “Software-defined radio using multi-core processor” by inventors Tan, et al., discloses a radio control board passing a plurality of digital samples between a memory of a computing device and a radio frequency (RF) transceiver coupled to a system bus of the computing device. Processing of the digital samples is carried out by one or more cores of a multi-core processor to implement a software-defined radio.

(13) U.S. Pat. No. 8,064,840 for “Method and system for determining spectrum availability within a network” by inventors McHenry, et al., filed Jun. 18, 2009, discloses an invention which determines spectrum holes for a communication network by accumulating the information obtained from previous received signals to determine the presence of a larger spectrum hole that allows a reduced listening period, higher transmit power and a reduced probability of interference with other networks and transmitters.

(14) U.S. Pat. No. 8,326,313 for “Method and system for dynamic spectrum access using detection periods” by inventors McHenry, et al., filed Aug. 14, 2009, discloses methods and systems for dynamic spectrum access (DSA) in a wireless network. A DSA-enabled device may sense spectrum use in a region and, based on the detected spectrum use, select one or more communication channels for use. The devices also may detect one or more other DSA-enabled devices with which they can form DSA networks. A DSA network may monitor spectrum use by cooperative and non-cooperative devices, to dynamically select one or more channels to use for communication while avoiding or reducing interference with other devices.

(15) U.S. Publication No. 2009/0143019 for “Method and apparatus for distributed spectrum sensing for wireless communication” by inventor Stephen J. Shellhammer, filed Jan. 4, 2008, discloses methods and apparatus for determining if a licensed signal having or exceeding a predetermined field strength is present in a wireless spectrum. The signal of interest maybe a television signal or a wireless microphone signal using licensed television spectrum.

(16) U.S. Publication No. 2013/0090071 for “Systems and methods for communication in a white space” by inventors Abraham, et al., filed Apr. 3, 2012, discloses systems, methods, and devices to communicate in a white space. In some aspects, wireless communication transmitted in the white space authorizes an initial transmission by a device. The wireless communication may include power information for determining a power at which to transmit the initial transmission. The initial transmission may be used to request information identifying one or more channels in the white space available for transmitting data.

(17) U.S. Publication No. 2012/0072986 for “Methods for detecting and classifying signals transmitted over a radio frequency spectrum” by inventors Livsics, et al., filed Nov. 1, 2011, discloses a method to classify a signal as non-cooperative (NC) or a target signal. The percentage of power above a first threshold is computed for a channel. Based on the percentage, a signal is classified as a narrowband signal. If the percentage indicates the absence of a narrowband signal, then a lower second threshold is applied to confirm the absence according to the percentage of power above the second threshold. The signal is classified as a narrowband signal or pre-classified as a wideband signal based on the percentage. Pre-classified wideband signals are classified as a

wideband NC signal ortarget signal using spectrum masks.

(18) U.S. Pat. No. 8,326,240 for “System for specific emitter identification” by inventors Kadambe, et al., filed Sep. 27, 2010, describes an apparatus for identifying a specific emitter in the presence of noise and/or interference including (a) a sensor configured to sense radio frequency signal and noise data, (b) a reference estimation unit configured to estimate a reference signal relating to the signal transmitted by one emitter, (c) a feature estimation unit configured to generate one or more estimates of one or more feature from the reference signal and the signal transmitted by that particular emitter, and (d) an emitter identifier configured to identify the signal transmitted by that particular emitter as belonging to a specific device (e.g., devices using Gaussian Mixture Models and the Bayesian decision engine). The apparatus may also include an SINR enhancement unit configured to enhance the SINR of the data before the reference estimation unit estimates the reference signal.

(19) U.S. Pat. No. 7,835,319 for “System and method for identifying wireless devices using pulse fingerprinting and sequence analysis” by inventor Sugar, filed May 9, 2007, discloses methods for identifying devices that are sources of wireless signals from received radio frequency (RF) energy, and, particularly, sources emitting frequency hopping spread spectrum (FHSS). Pulse metric data is generated from the received RF energy and represents characteristics associated thereto. The pulses are partitioned into groups based on their pulse metric data such that a group comprises pulses having similarities for at least one item of pulse metric data. Sources of the wireless signals are identified based on the partitioning process. The partitioning process involves iteratively subdividing each group into subgroups until all resulting subgroups contain pulses determined to be from a single source. At each iteration, subdividing is performed based on different pulse metric data than at a prior iteration. Ultimately, output data is generated (e.g., a device name for display) that identifies a source of wireless signals for any subgroup that is determined to contain pulses from a single source.

(20) U.S. Pat. No. 8,131,239 for “Method and apparatus for remote detection of radio-frequency devices” by inventors Walker, et al., filed Aug. 21, 2007, describes methods and apparatus for detecting the presence of electronic communications devices, such as cellular phones, including a complex RF stimulus is transmitted into a target area, and nonlinear reflection signals received from the target area are processed to obtain a response measurement. The response measurement is compared to a pre-determined filter response profile to detect the presence of a radio device having a corresponding filter response characteristic. In some embodiments, the pre-determined filter response profile comprises a pre-determined band-edge profile, so that comparing the response measurement to a pre-determined filter response profile comprises comparing the response measurement to the pre-determined band-edge profile to detect the presence of a radio device having a corresponding band-edge characteristic. Invention aims to be useful in detecting hidden electronic devices.

(21) U.S. Pat. No. 8,369,305 for “Correlating multiple detections of wireless devices without a unique identifier” by inventors Diener, et al., filed Jun. 30, 2008, describes at a plurality of first devices, wireless transmissions are received at different locations in a region where multiple target devices may be emitting, and identifier data is subsequently generated. Similar identifier data associated with received emissions at multiple first devices are grouped together into a cluster record that potentially represents the same target device detected by multiple first devices. Data is stored that represents a plurality of cluster records from identifier data associated with received emissions made over time by multiple first devices. The cluster records are analyzed over time to correlate detections of target devices across multiple first devices. It aims to lessen disruptions caused by devices using the same frequency and to protect data.

(22) U.S. Pat. No. 8,155,649 for “Method and system for classifying communication signals in a dynamic spectrum access system” by inventors McHenry, et al., filed Aug. 14, 2009, discloses methods and systems for dynamic spectrum access (DSA) in a wireless network wherein a DSA-

enabled device may sense spectrum use in a region and, based on the detected spectrum use, select one or more communication channels for use. The devices also may detect one or more other DSA-enabled devices with which they can form DSA networks. A DSA network may monitor spectrum use by cooperative and non-cooperative devices, to dynamically select one or more channels to use for communication while avoiding or reducing interference with other devices. A DSA network may include detectors such as a narrow-band detector, wide-band detector, TV detector, radar detector, a wireless microphone detector, or any combination thereof.

(23) U.S. Pat. No. 8,494,464 for “Cognitive networked electronic warfare” by inventors Kadambe, et al., filed Sep. 8, 2010, describes an apparatus for sensing and classifying radio communications including sensor units configured to detect RF signals, a signal classifier configured to classify the detected RF signals into a classification, the classification including at least one known signal type and an unknown signal type, a clustering learning algorithm capable of finding clusters of common signals among the previously seen unknown signals; it is then further configured to use these clusters to retrain the signal classifier to recognize these signals as a new signal type, aiming to provide signal identification to better enable electronic attacks and jamming signals.

(24) U.S. Publication No. 2011/0059747 for “Sensing Wireless Transmissions From a Licensed User of a Licensed Spectral Resource” by inventors Lindoff, et al., filed Sep. 7, 2009, describes sensing wireless transmissions from a licensed user of a licensed spectral resource includes obtaining information indicating a number of adjacent sensors that are concurrently sensing wireless transmissions from the licensed user of the licensed spectral resource. Such information can be obtained from a main node controlling the sensor and its adjacent sensors, or by the sensor itself (e.g., by means of short-range communication equipment targeting any such adjacent sensors). A sensing rate is then determined as a function, at least in part, of the information indicating the number of adjacent sensors that are concurrently sensing wireless transmissions from the licensed user of the licensed spectral resource. Receiver equipment is then periodically operated at the determined sensing rate, wherein the receiver equipment is configured to detect wireless transmissions from the licensed user of the licensed spectral resource.

(25) U.S. Pat. No. 8,463,195 for “Methods and apparatus for spectrum sensing of signal features in a wireless channel” by inventor Shellhammer, filed Nov. 13, 2009, discloses methods and apparatus for sensing features of a signal in a wireless communication system are disclosed. The disclosed methods and apparatus sense signal features by determining a number of spectral density estimates, where each estimate is derived based on reception of the signal by a respective antenna in a system with multiple sensing antennas. The spectral density estimates are then combined, and the signal features are sensed based on the combination of the spectral density estimates. Invention aims to increase sensing performance by addressing problems associated with Rayleigh fading, which causes signals to be less detectable.

(26) U.S. Pat. No. 8,151,311 for “System and method of detecting potential video traffic interference” by inventors Huffman, et al., filed Nov. 30, 2007, describes a method of detecting potential video traffic interference at a video head-end of a video distribution network is disclosed and includes detecting, at a video head-end, a signal populating an ultra-high frequency (UHF) white space frequency. The method also includes determining that a strength of the signal is equal to or greater than a threshold signal strength. Further, the method includes sending an alert from the video head-end to a network management system. The alert indicates that the UHF white space frequency is populated by a signal having a potential to interfere with video traffic delivered via the video head-end. Cognitive radio technology, various sensing mechanisms (energy sensing, National Television System Committee signal sensing, Advanced Television Systems Committee sensing), filtering, and signal reconstruction are disclosed.

(27) U.S. Pat. No. 8,311,509 for “Detection, communication and control in multimode cellular, TDMA, GSM, spread spectrum, CDMA, OFDM, WiLAN, and WiFi systems” by inventor Feher, filed Oct. 31, 2007, teaches a device for detection of signals, with location finder or location tracker

or navigation signal and with Modulation Demodulation (Modem) Format Selectable (MFS) communication signal. Processor for processing a digital signal into cross-correlated in-phase and quadrature-phase filtered signal and for processing a voice signal into Orthogonal Frequency Division Multiplexed (OFDM) or Orthogonal Frequency Division Multiple Access (OFDMA) signal. Each is used in a Wireless Local Area Network (WLAN) and in Voice over Internet Protocol (VOIP) network. Device and location finder with Time Division Multiple Access (TDMA), Global Mobile System (GSM) and spread spectrum Code Division Multiple Access (CDMA) is used in a cellular network. Polar and quadrature modulator and two antenna transmitter for transmission of provided processed signal. Transmitter with two amplifiers operated in separate radio frequency (RF) bands. One transmitter is operated as a Non-Linearly Amplified (NLA) transmitter and the other transmitter is operated as a linearly amplified or linearized amplifier transmitter.

(28) U.S. Pat. No. 8,514,729 for “Method and system for analyzing RF signals in order to detect and classify actively transmitting RF devices” by inventor Blackwell, filed Apr. 3, 2009, discloses methods and apparatuses to analyze RF signals in order to detect and classify RF devices in wireless networks are described. The method includes detecting one or more radio frequency (RF) samples; determining burst data by identifying start and stop points of the one or more RF samples; comparing time domain values for an individual burst with time domain values of one or more predetermined RF device profiles; generating a human-readable result indicating whether the individual burst should be assigned to one of the predetermined RF device profiles; and, classifying the individual burst if assigned to one of the predetermined RF device profiles as being a WiFi device or a non-WiFi device with the non-WiFi device being a RF interference source to a wireless network.

(29) However, none of the prior art references provide solutions to the limitations and longstanding unmet needs existing in this area for automatically identifying open space in a wireless communications spectrum. Thus, there remains a need for automated identification of open space in a wireless communications spectrum in near real-time.

SUMMARY OF THE INVENTION

(30) The present invention addresses the longstanding, unmet needs existing in the prior art and commercial sectors to provide solutions to the at least four major problems existing before the present invention. The present invention relates to systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, data types carried by the signals, and estimated signal origins may be identified.

(31) It is an object of this invention is to provide an apparatus for identifying signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device, and then to identify open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

(32) The present invention further provides systems for identifying white space in wireless communications spectrum by detecting and analyzing signals from any signal emitting devices including at least one apparatus, wherein the at least one apparatus is operable for network-based communication with at least one server computer including a database, and/or with at least one other apparatus, but does not require a connection to the at least one server computer to be operable for identifying signal emitting devices; wherein each of the apparatus is operable for identifying

signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device, and then to identify open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

(33) The present invention is further directed to a method for identifying open space in a wireless communications spectrum including the steps of: providing a device for measuring characteristics of signals from signal emitting devices in a spectrum associated with wireless communications, with measured data characteristics including frequency, power, bandwidth, duration, modulation, and combinations thereof; the device including a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals within the spectrum; and further including the following steps performed within the device housing: assessing whether the measured data includes analog and/or digital signal(s); determining a best fit based on frequency, if the measured power spectrum is designated in an historical or a reference database(s) for frequency ranges; automatically determining a category for either analog or digital signals, based on power and sideband combined with frequency allocation; determining a TDM/FDM/CDM signal, based on duration and bandwidth; identifying at least one signal emitting device from the composite results of the foregoing steps; and then automatically identifying the open space available for wireless communications, based upon the information about the signal emitting device(s) operating in the predetermined spectrum.

(34) Additionally, the present invention provides systems, apparatus, and methods for identifying open space in a wireless communications spectrum using an apparatus having a multiplicity of processors and memory, sensors, and communications transmitters and receivers, all constructed and configured within a housing for automated analysis of detected signals from signal emitting devices, determination of signal duration and other signal characteristics, and automatically generating information relating to open space within the spectrum for wireless communication.

(35) These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments of the invention, and together with the general description given above and the detailed description given below, serve to explain the features of the invention.

(2) FIG. 1 is a system block diagram of a wireless environment suitable for use with the various embodiments.

(3) FIG. 2A is a block diagram of a spectrum management device according to an embodiment.

(4) FIG. 2B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment.

(5) FIG. 3 is a process flow diagram illustrating an embodiment method for identifying a signal.

(6) FIG. 4 is a process flow diagram illustrating an embodiment method for measuring sample blocks of a radio frequency scan.

(7) FIGS. 5A-5C are a process flow diagram illustrating an embodiment method for determining

signal parameters.

(8) FIG. 6 is a process flow diagram illustrating an embodiment method for displaying signal identifications.

(9) FIG. 7 is a process flow diagram illustrating an embodiment method for displaying one or more open frequency.

(10) FIG. 8A is a block diagram of a spectrum management device according to another embodiment.

(11) FIG. 8B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to another embodiment.

(12) FIG. 9 is a process flow diagram illustrating an embodiment method for determining protocol data and symbol timing data.

(13) FIG. 10 is a process flow diagram illustrating an embodiment method for calculating signal degradation data.

(14) FIG. 11 is a process flow diagram illustrating an embodiment method for displaying signal and protocol identification information.

(15) FIG. 12A is a block diagram of a spectrum management device according to a further embodiment.

(16) FIG. 12B is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to a further embodiment.

(17) FIG. 13 is a process flow diagram illustrating an embodiment method for estimating a signal origin based on a frequency difference of arrival.

(18) FIG. 14 is a process flow diagram illustrating an embodiment method for displaying an indication of an identified data type within a signal.

(19) FIG. 15 is a process flow diagram illustrating an embodiment method for determining modulation type, protocol data, and symbol timing data.

(20) FIG. 16 is a process flow diagram illustrating an embodiment method for tracking a signal origin.

(21) FIG. 17 is a schematic diagram illustrating an embodiment for scanning and finding open space.

(22) FIG. 18 is a diagram of an embodiment wherein software defined radio nodes are in communication with a master transmitter and device sensing master.

(23) FIG. 19 is a process flow diagram of an embodiment method of temporally dividing up data into intervals for power usage analysis.

(24) FIG. 20 is a flow diagram illustrating an embodiment wherein frequency to license matching occurs.

(25) FIG. 21 is a flow diagram illustrating an embodiment method for reporting power usage information.

(26) FIG. 22 is a flow diagram illustrating an embodiment method for creating frequency arrays.

(27) FIG. 23 is a flow diagram illustrating an embodiment method for reframe and aggregating power when producing frequency arrays.

(28) FIG. 24 is a flow diagram illustrating an embodiment method of reporting license expirations.

(29) FIG. 25 is a flow diagram illustrating an embodiment method of reporting frequency power use.

(30) FIG. 26 is a flow diagram illustrating an embodiment method of connecting devices.

(31) FIG. 27 is a flow diagram illustrating an embodiment method of addressing collisions.

(32) FIG. 28 is a schematic diagram of an embodiment of the invention illustrating a virtualized computing network and a plurality of distributed devices.

DETAILED DESCRIPTION

(33) Priority U.S. Patent Applications 61/789,758, Ser. Nos. 13/912,683, 13/912,893, 13/913,013, 14/028,873, 14/082,916, 14/082,930, 14/086,871, 14/273,157 and 14/329,838 each are herein

incorporated by reference in their entirety.

(34) Referring now to the drawings in general, the illustrations are for the purpose of describing at least one preferred embodiment and/or examples of the invention and are not intended to limit the invention thereto. Various embodiments are described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers are used throughout the drawings to refer to the same or like parts. References made to particular examples and implementations are for illustrative purposes, and are not intended to limit the scope of the invention or the claims.

(35) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

(36) The present invention provides systems, methods, and devices for spectrum analysis and management by identifying, classifying, and cataloging at least one or a multiplicity of signals of interest based on radio frequency measurements and location and other measurements, and using near real-time parallel processing of signals and their corresponding parameters and characteristics in the context of historical and static data for a given spectrum.

(37) The systems, methods and apparatus according to the present invention preferably have the ability to detect in near real-time, and more preferably to detect, sense, measure, and/or analyze in near real-time, and more preferably to perform any near real-time operations within about 1 second or less. Advantageously, the present invention and its real-time functionality described herein uniquely provide and enable the apparatus units to compare to historical data, to update data and/or information, and/or to provide more data and/or information on the open space, on the device that may be occupying the open space, and combinations, in the near real-time compared with the historically scanned (15 min to 30 days) data, or historical database information.

(38) The systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, data types carried by the signals, and estimated signal origins may be identified.

(39) Embodiments are directed to a spectrum management device that may be configurable to obtain spectrum data over a wide range of wireless communication protocols. Embodiments may also provide for the ability to acquire data from and sending data to database depositories that may be used by a plurality of spectrum management customers.

(40) In one embodiment, a spectrum management device may include a signal spectrum analyzer that may be coupled with a database system and spectrum management interface. The device may be portable or may be a stationary installation and may be updated with data to allow the device to manage different spectrum information based on frequency, bandwidth, signal power, time, and location of signal propagation, as well as modulation type and format and to provide signal identification, classification, and geo-location. A processor may enable the device to process spectrum power density data as received and to process raw I/Q complex data that may be used for further signal processing, signal identification, and data extraction.

(41) In an embodiment, a spectrum management device may comprise a low noise amplifier that receives a radio frequency (RF) energy from an antenna. The antenna may be any antenna structure that is capable of receiving RF energy in a spectrum of interest. The low noise amplifier may filter and amplify the RF energy. The RF energy may be provided to an RF translator. The RF translator may perform a fast Fourier transform (FFT) and either a square magnitude or a fast convolution spectral periodogram function to convert the RF measurements into a spectral representation. In an embodiment, the RF translator may also store a timestamp to facilitate calculation of a time of arrival and an angle of arrival. The In-Phase and Quadrature (I/Q) data may be provided to a spectral analysis receiver or it may be provided to a sample data store where it may be stored

without being processed by a spectral analysis receiver. The input RF energy may also be directly digital down-converted and sampled by an analog to digital converter (ADC) to generate complex I/Q data. The complex I/Q data may be equalized to remove multipath, fading, white noise and interference from other signaling systems by fast parallel adaptive filter processes. This data may then be used to calculate modulation type and baud rate. Complex sampled I/Q data may also be used to measure the signal angle of arrival and time of arrival. Such information as angle of arrival and time of arrival may be used to compute more complex and precise direction finding. In addition, they may be used to apply geo-location techniques. Data may be collected from known signals or unknown signals and time spaced in order to provide expedient information. I/Q sampled data may contain raw signal data that may be used to demodulate and translate signals by streaming them to a signal analyzer or to a real-time demodulator software defined radio that may have the newly identified signal parameters for the signal of interest. The inherent nature of the input RF allows for any type of signal to be analyzed and demodulated based on the reconfiguration of the software defined radio interfaces.

(42) A spectral analysis receiver may be configured to read raw In-Phase (I) and Quadrature (Q) data and either translate directly to spectral data or down convert to an intermediate frequency (IF) up to half the Nyquist sampling rate to analyze the incoming bandwidth of a signal. The translated spectral data may include measured values of signal energy, frequency, and time. The measured values provide attributes of the signal under review that may confirm the detection of a particular signal of interest within a spectrum of interest. In an embodiment, a spectral analysis receiver may have a referenced spectrum input of 0 Hz to 12.4 GHz with capability of fiber optic input for spectrum input up to 60 GHz.

(43) In an embodiment, the spectral analysis receiver may be configured to sample the input RF data by fast analog down-conversion of the RF signal. The down-converted signal may then be digitally converted and processed by fast convolution filters to obtain a power spectrum. This process may also provide spectrum measurements including the signal power, the bandwidth, the center frequency of the signal as well as a Time of Arrival (TOA) measurement. The TOA measurement may be used to create a timestamp of the detected signal and/or to generate a time difference of arrival iterative process for direction finding and fast triangulation of signals. In an embodiment, the sample data may be provided to a spectrum analysis module. In an embodiment, the spectrum analysis module may evaluate the sample data to obtain the spectral components of the signal.

(44) In an embodiment, the spectral components of the signal may be obtained by the spectrum analysis module from the raw I/Q data as provided by an RF translator. The I/Q data analysis performed by the spectrum analysis module may operate to extract more detailed information about the signal, including by way of example, modulation type (e.g., FM, AM, QPSK, 16QAM, etc.) and/or protocol (e.g., GSM, CDMA, OFDM, LTE, etc.). In an embodiment, the spectrum analysis module may be configured by a user to obtain specific information about a signal of interest. In an alternate embodiment, the spectral components of the signal may be obtained from power spectral component data produced by the spectral analysis receiver.

(45) In an embodiment, the spectrum analysis module may provide the spectral components of the signal to a data extraction module. The data extraction module may provide the classification and categorization of signals detected in the RF spectrum. The data extraction module may also acquire additional information regarding the signal from the spectral components of the signal. For example, the data extraction module may provide modulation type, bandwidth, and possible system in use information. In another embodiment, the data extraction module may select and organize the extracted spectral components in a format selected by a user.

(46) The information from the data extraction module may be provided to a spectrum management module. The spectrum management module may generate a query to a static database to classify a signal based on its components. For example, the information stored in static database may be used

to determine the spectral density, center frequency, bandwidth, baud rate, modulation type, protocol (e.g., GSM, CDMA, OFDM, LTE, etc.), system or carrier using licensed spectrum, location of the signal source, and a timestamp of the signal of interest. These data points may be provided to a data store for export. In an embodiment and as more fully described below, the data store may be configured to access mapping software to provide the user with information on the location of the transmission source of the signal of interest. In an embodiment, the static database includes frequency information gathered from various sources including, but not limited to, the Federal Communication Commission, the International Telecommunication Union, and data from users. As an example, the static database may be an SQL database. The data store may be updated, downloaded or merged with other devices or with its main relational database. Software API applications may be included to allow database merging with third-party spectrum databases that may only be accessed securely.

(47) In the various embodiments, the spectrum management device may be configured in different ways. In an embodiment, the front end of system may comprise various hardware receivers that may provide In-Phase and Quadrature complex data. The front end receiver may include API set commands via which the system software may be configured to interface (i.e., communicate) with a third party receiver. In an embodiment, the front end receiver may perform the spectral computations using FFT (Fast Fourier Transform) and other DSP (Digital Signal Processing) to generate a fast convolution periodogram that may be re-sampled and averaged to quickly compute the spectral density of the RF environment.

(48) In an embodiment, cyclic processes may be used to average and correlate signal information by extracting the changes inside the signal to better identify the signal of interest that is present in the RF space. A combination of amplitude and frequency changes may be measured and averaged over the bandwidth time to compute the modulation type and other internal changes, such as changes in frequency offsets, orthogonal frequency division modulation, changes in time (e.g., Time Division Multiplexing), and/or changes in I/Q phase rotation used to compute the baud rate and the modulation type. In an embodiment, the spectrum management device may have the ability to compute several processes in parallel by use of a multi-core processor and along with several embedded field programmable gate arrays (FPGA). Such multi-core processing may allow the system to quickly analyze several signal parameters in the RF environment at one time in order to reduce the amount of time it takes to process the signals. The amount of signals computed at once may be determined by their bandwidth requirements. Thus, the capability of the system may be based on a maximum frequency $F_s/2$. The number of signals to be processed may be allocated based on their respective bandwidths. In another embodiment, the signal spectrum may be measured to determine its power density, center frequency, bandwidth and location from which the signal is emanating and a best match may be determined based on the signal parameters based on information criteria of the frequency.

(49) In another embodiment, a GPS and direction finding location (DF) system may be incorporated into the spectrum management device and/or available to the spectrum management device. Adding GPS and DF ability may enable the user to provide a location vector using the National Marine Electronics Association's (NMEA) standard form. In an embodiment, location functionality is incorporated into a specific type of GPS unit, such as a U.S. government issued receiver. The information may be derived from the location presented by the database internal to the device, a database imported into the device, or by the user inputting geo-location parameters of longitude and latitude which may be derived as degrees, minutes and seconds, decimal minutes, or decimal form and translated to the necessary format with the default being 'decimal' form. This functionality may be incorporated into a GPS unit. The signal information and the signal classification may then be used to locate the signaling device as well as to provide a direction finding capability.

(50) A type of triangulation using three units as a group antenna configuration performs direction

finding by using multilateration. Commonly used in civil and military surveillance applications, multilateration is able to accurately locate an aircraft, vehicle, or stationary emitter by measuring the “Time Difference of Arrival” (TDOA) of a signal from the emitter at three or more receiver sites. If a pulse is emitted from a platform, it will arrive at slightly different times at two spatially separated receiver sites, the TDOA being due to the different distances of each receiver from the platform. This location information may then be supplied to a mapping process that utilizes a database of mapping images that are extracted from the database based on the latitude and longitude provided by the geo-location or direction finding device. The mapping images may be scanned in to show the points of interest where a signal is either expected to be emanating from based on the database information or from an average taken from the database information and the geo-location calculation performed prior to the mapping software being called. The user can control the map to maximize or minimize the mapping screen to get a better view which is more fit to provide information of the signal transmissions. In an embodiment, the mapping process does not rely on outside mapping software. The mapping capability has the ability to generate the map image and to populate a mapping database that may include information from third party maps to meet specific user requirements.

(51) In an embodiment, triangulation and multilateration may utilize a Bayesian type filter that may predict possible movement and future location and operation of devices based on input collected from the TDOA and geolocation processes and the variables from the static database pertaining to the specified signal of interest. The Bayesian filter takes the input changes in time difference and its inverse function (i.e., frequency difference) and takes an average changes in signal variation to detect and predict the movement of the signals. The signal changes are measured within 1 ns time difference and the filter may also adapt its gradient error calculation to remove unwanted signals that may cause errors due to signal multipath, inter-symbol interference, and other signal noise.

(52) In an embodiment the changes within a 1 ns time difference for each sample for each unique signal may be recorded. The spectrum management device may then perform the inverse and compute and record the frequency difference and phase difference between each sample for each unique signal. The spectrum management device may take the same signal and calculates an error based on other input signals coming in within the 1 ns time and may average and filter out the computed error to equalize the signal. The spectrum management device may determine the time difference and frequency difference of arrival for that signal and compute the odds of where the signal is emanating from based on the frequency band parameters presented from the spectral analysis and processor computations, and determines the best position from which the signal is transmitted (i.e., origin of the signal).

(53) FIG. 1 illustrates a wireless environment **100** suitable for use with the various embodiments. The wireless environment **100** may include various sources **104**, **106**, **108**, **110**, **112**, and **114** generating various radio frequency (RF) signals **116**, **118**, **120**, **122**, **124**, **126**. As an example, mobile devices **104** may generate cellular RF signals **116**, such as CDMA, GSM, 3G signals, etc. As another example, wireless access devices **106**, such as Wi-Fi® routers, may generate RF signals **118**, such as Wi-Fi® signals. As a further example, satellites **108**, such as communication satellites or GPS satellites, may generate RF signals **120**, such as satellite radio, television, or GPS signals. As a still further example, base stations **110**, such as a cellular base station, may generate RF signals **122**, such as CDMA, GSM, 3G signals, etc. As another example, radio towers **112**, such as local AM or FM radio stations, may generate RF signals **124**, such as AM or FM radio signals. As another example, government service provides **114**, such as police units, fire fighters, military units, air traffic control towers, etc. may generate RF signals **126**, such as radio communications, tracking signals, etc. The various RF signals **116**, **118**, **120**, **122**, **124**, **126** may be generated at different frequencies, power levels, in different protocols, with different modulations, and at different times. The various sources **104**, **106**, **108**, **110**, **112**, and **114** may be assigned frequency bands, power limitations, or other restrictions, requirements, and/or licenses by a government

spectrum control entity, such as the FCC. However, with so many different sources **104**, **106**, **108**, **110**, **112**, and **114** generating so many different RF signals **116**, **118**, **120**, **122**, **124**, **126**, overlaps, interference, and/or other problems may occur. A spectrum management device **102** in the wireless environment **100** may measure the RF energy in the wireless environment **100** across a wide spectrum and identify the different RF signals **116**, **118**, **120**, **122**, **124**, **126** which may be present in the wireless environment **100**. The identification and cataloging of the different RF signals **116**, **118**, **120**, **122**, **124**, **126** which may be present in the wireless environment **100** may enable the spectrum management device **102** to determine available frequencies for use in the wireless environment **100**. In addition, the spectrum management device **102** may be able to determine if there are available frequencies for use in the wireless environment **100** under certain conditions (i.e., day of week, time of day, power level, frequency band, etc.). In this manner, the RF spectrum in the wireless environment **100** may be managed.

(54) FIG. 2A is a block diagram of a spectrum management device **202** according to an embodiment. The spectrum management device **202** may include an antenna structure **204** configured to receive RF energy expressed in a wireless environment. The antenna structure **204** may be any type antenna, and may be configured to optimize the receipt of RF energy across a wide frequency spectrum. The antenna structure **204** may be connected to one or more optional amplifiers and/or filters **208** which may boost, smooth, and/or filter the RF energy received by antenna structure **204** before the RF energy is passed to an RF receiver **210** connected to the antenna structure **204**. In an embodiment, the RF receiver **210** may be configured to measure the RF energy received from the antenna structure **204** and/or optional amplifiers and/or filters **208**. In an embodiment, the RF receiver **210** may be configured to measure RF energy in the time domain and may convert the RF energy measurements to the frequency domain. In an embodiment, the RF receiver **210** may be configured to generate spectral representation data of the received RF energy. The RF receiver **210** may be any type RF receiver, and may be configured to generate RF energy measurements over a range of frequencies, such as 0 kHz to 24 GHz, 9 kHz to 6 GHz, etc. In an embodiment, the frequency scanned by the RF receiver **210** may be user selectable. In an embodiment, the RF receiver **210** may be connected to a signal processor **214** and may be configured to output RF energy measurements to the signal processor **214**. As an example, the RF receiver **210** may output raw In-Phase (I) and Quadrature (Q) data to the signal processor **214**. As another example, the RF receiver **210** may apply signals processing techniques to output complex In-Phase (I) and Quadrature (Q) data to the signal processor **214**. In an embodiment, the spectrum management device may also include an antenna **206** connected to a location receiver **212**, such as a GPS receiver, which may be connected to the signal processor **214**. The location receiver **212** may provide location inputs to the signal processor **214**.

(55) The signal processor **214** may include a signal detection module **216**, a comparison module **222**, a timing module **224**, and a location module **225**. Additionally, the signal processor **214** may include an optional memory module **226** which may include one or more optional buffers **228** for storing data generated by the other modules of the signal processor **214**.

(56) In an embodiment, the signal detection module **216** may operate to identify signals based on the RF energy measurements received from the RF receiver **210**. The signal detection module **216** may include a Fast Fourier Transform (FFT) module **217** which may convert the received RF energy measurements into spectral representation data. The signal detection module **216** may include an analysis module **221** which may analyze the spectral representation data to identify one or more signals above a power threshold. A power module **220** of the signal detection module **216** may control the power threshold at which signals may be identified. In an embodiment, the power threshold may be a default power setting or may be a user selectable power setting. A noise module **219** of the signal detection module **216** may control a signal threshold, such as a noise threshold, at or above which signals may be identified. The signal detection module **216** may include a parameter module **218** which may determine one or more signal parameters for any identified

signals, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, etc. In an embodiment, the signal processor **214** may include a timing module **224** which may record time information and provide the time information to the signal detection module **216**. Additionally, the signal processor **214** may include a location module **225** which may receive location inputs from the location receiver **212** and determine a location of the spectrum management device **202**. The location of the spectrum management device **202** may be provided to the signal detection module **216**.

(57) In an embodiment, the signal processor **214** may be connected to one or more memory **230**. The memory **230** may include multiple databases, such as a history or historical database **232** and characteristics listing **236**, and one or more buffers **240** storing data generated by signal processor **214**. While illustrated as connected to the signal processor **214** the memory **230** may also be on chip memory residing on the signal processor **214** itself. In an embodiment, the history or historical database **232** may include measured signal data **234** for signals that have been previously identified by the spectrum management device **202**. The measured signal data **234** may include the raw RF energy measurements, time stamps, location information, one or more signal parameters for any identified signals, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, etc., and identifying information determined from the characteristics listing **236**. In an embodiment, the history or historical database **232** may be updated as signals are identified by the spectrum management device **202**. In an embodiment, the characteristic listing **236** may be a database of static signal data **238**. The static signal data **238** may include data gathered from various sources including by way of example and not by way of limitation the Federal Communication Commission, the International Telecommunication Union, telecom providers, manufacture data, and data from spectrum management device users. Static signal data **238** may include known signal parameters of transmitting devices, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, geographic information for transmitting devices, and any other data that may be useful in identifying a signal. In an embodiment, the static signal data **238** and the characteristic listing **236** may correlate signal parameters and signal identifications. As an example, the static signal data **238** and characteristic listing **236** may list the parameters of the local fire and emergency communication channel correlated with a signal identification indicating that signal is the local fire and emergency communication channel.

(58) In an embodiment, the signal processor **214** may include a comparison module **222** which may match data generated by the signal detection module **216** with data in the history or historical database **232** and/or characteristic listing **236**. In an embodiment the comparison module **222** may receive signal parameters from the signal detection module **216**, such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration, and/or receive parameter from the timing module **224** and/or location module **225**. The parameter match module **223** may retrieve data from the history or historical database **232** and/or the characteristic listing **236** and compare the retrieved data to any received parameters to identify matches. Based on the matches the comparison module may identify the signal. In an embodiment, the signal processor **214** may be optionally connected to a display **242**, an input device **244**, and/or network transceiver **246**. The display **242** may be controlled by the signal processor **214** to output spectral representations of received signals, signal characteristic information, and/or indications of signal identifications on the display **242**. In an embodiment, the input device **244** may be any input device, such as a keyboard and/or knob, mouse, virtual keyboard or even voice recognition, enabling the user of the spectrum management device **202** to input information for use by the signal processor **214**. In an embodiment, the network transceiver **246** may enable the spectrum management device **202** to exchange data with wired and/or wireless networks, such as to update the characteristic listing **236** and/or upload information from the history or historical database **232**.

(59) FIG. 2B is a schematic logic flow diagram illustrating logical operations which may be performed by a spectrum management device **202** according to an embodiment. A receiver **210** may output RF energy measurements, such as I and Q data to a FFT module **252** which may generate a spectral representation of the RF energy measurements which may be output on a display **242**. The I and Q data may also be buffered in a buffer **256** and sent to a signal detection module **216**. The signal detection module **216** may receive location inputs from a location receiver **212** and use the received I and Q data to detect signals. Data from the signal detection module **216** may be buffered in a buffer **262** and written into a history or historical database **232**. Additionally, data from the historical database may be used to aid in the detection of signals by the signal detection module **216**. The signal parameters of the detected signals may be determined by a signal parameters module **218** using information from the history or historical database **232** and/or a static database **238** listing signal characteristics. Data from the signal parameters module **218** may be buffered in a buffer **268**, stored in the history or historical database **232**, and/or sent to the signal detection module **216** and/or display **242**. In this manner, signals may be detected and indications of the signal identification may be displayed to a user of the spectrum management device.

(60) FIG. 3 illustrates a process flow of an embodiment method **300** for identifying a signal. In an embodiment the operations of method **300** may be performed by the processor **214** of a spectrum management device **202**. In block **302** the processor **214** may determine the location of the spectrum management device **202**. In an embodiment, the processor **214** may determine the location of the spectrum management device **202** based on a location input, such as GPS coordinates, received from a location receiver, such as a GPS receiver **212**. In block **304** the processor **214** may determine the time. As an example, the time may be the current clock time as determined by the processor **214** and may be a time associated with receiving RF measurements. In block **306** the processor **214** may receive RF energy measurements. In an embodiment, the processor **214** may receive RF energy measurements from an RF receiver **210**. In block **308** the processor **214** may convert the RF energy measurements to spectral representation data. As an example, the processor may apply a Fast Fourier Transform (FFT) to the RF energy measurements to convert them to spectral representation data. In optional block **310** the processor **214** may display the spectral representation data on a display **242** of the spectrum management device **202**, such as in a graph illustrating amplitudes across a frequency spectrum.

(61) In block **312** the processor **214** may identify one or more signal above a threshold. In an embodiment, the processor **214** may analyze the spectral representation data to identify a signal above a power threshold. A power threshold may be an amplitude measure selected to distinguish RF energies associated with actual signals from noise. In an embodiment, the power threshold may be a default value. In another embodiment, the power threshold may be a user selectable value. In block **314** the processor **214** may determine signal parameters of any identified signal or signals of interest. As examples, the processor **214** may determine signal parameters such as center frequency, bandwidth, power, number of detected signals, frequency peak, peak power, average power, signal duration for the identified signals. In block **316** the processor **214** may store the signal parameters of each identified signal, a location indication, and time indication for each identified signal in a history database **232**. In an embodiment, a history database **232** may be a database resident in a memory **230** of the spectrum management device **202** which may include data associated with signals actually identified by the spectrum management device.

(62) In block **318** the processor **214** may compare the signal parameters of each identified signal to signal parameters in a signal characteristic listing. In an embodiment, the signal characteristic listing may be a static database **238** stored in the memory **230** of the spectrum management device **202** which may correlate signal parameters and signal identifications. In determination block **320** the processor **214** may determine whether the signal parameters of the identified signal or signals match signal parameters in the characteristic listing **236**. In an embodiment, a match may be determined based on the signal parameters being within a specified tolerance of one another. As an

example, a center frequency match may be determined when the center frequencies are within plus or minus 1 kHz of each other. In this manner, differences between real world measured conditions of an identified signal and ideal conditions listed in a characteristics listing may be accounted for in identifying matches. If the signal parameters do not match (i.e., determination block **320**="No"), in block **326** the processor **214** may display an indication that the signal is unidentified on a display **242** of the spectrum management device **202**. In this manner, the user of the spectrum management device may be notified that a signal is detected, but has not been positively identified. If the signal parameters do match (i.e., determination block **320**="Yes"), in block **324** the processor **214** may display an indication of the signal identification on the display **242**. In an embodiment, the signal identification displayed may be the signal identification correlated to the signal parameter in the signal characteristic listing which matched the signal parameter for the identified signal. Upon displaying the indications in blocks **324** or **326** the processor **214** may return to block **302** and cyclically measure and identify further signals of interest.

(63) FIG. 4 illustrates an embodiment method **400** for measuring sample blocks of a radio frequency scan. In an embodiment the operations of method **400** may be performed by the processor **214** of a spectrum management device **202**. As discussed above, in blocks **306** and **308** the processor **214** may receive RF energy measurements and convert the RF energy measurements to spectral representation data. In block **402** the processor **214** may determine a frequency range at which to sample the RF spectrum for signals of interest. In an embodiment, a frequency range may be a frequency range of each sample block to be analyzed for potential signals. As an example, the frequency range may be 240 kHz. In an embodiment, the frequency range may be a default value. In another embodiment, the frequency range may be a user selectable value. In block **404** the processor **214** may determine a number (N) of sample blocks to measure. In an embodiment, each sample block may be sized to the determined of default frequency range, and the number of sample blocks may be determined by dividing the spectrum of the measured RF energy by the frequency range. In block **406** the processor **214** may assign each sample block a respective frequency range. As an example, if the determined frequency range is 240 kHz, the first sample block may be assigned a frequency range from 0 kHz to 240 kHz, the second sample block may be assigned a frequency range from 240 kHz to 480 kHz, etc. In block **408** the processor **214** may set the lowest frequency range sample block as the current sample block. In block **409** the processor **214** may measure the amplitude across the set frequency range for the current sample block. As an example, at each frequency interval (such as 1 Hz) within the frequency range of the sample block the processor **214** may measure the received signal amplitude. In block **410** the processor **214** may store the amplitude measurements and corresponding frequencies for the current sample block. In determination block **414** the processor **214** may determine if all sample blocks have been measured. If all sample blocks have not been measured (i.e., determination block **414**="No"), in block **416** the processor **214** may set the next highest frequency range sample block as the current sample block. As discussed above, in blocks **409**, **410**, and **414** the processor **214** may measure and store amplitudes and determine whether all blocks are sampled. If all blocks have been sampled (i.e., determination block **414**="Yes"), the processor **214** may return to block **306** and cyclically measure further sample blocks.

(64) FIGS. 5A, 5B, and 5C illustrate the process flow for an embodiment method **500** for determining signal parameters. In an embodiment the operations of method **500** may be performed by the processor **214** of a spectrum management device **202**. Referring to FIG. 5A, in block **502** the processor **214** may receive a noise floor average setting. In an embodiment, the noise floor average setting may be an average noise level for the environment in which the spectrum management device **202** is operating. In an embodiment, the noise floor average setting may be a default setting and/or may be user selectable setting. In block **504** the processor **214** may receive the signal power threshold setting. In an embodiment, the signal power threshold setting may be an amplitude measure selected to distinguish RF energies associated with actual signals from noise. In an

embodiment the signal power threshold may be a default value and/or may be a user selectable setting. In block **506** the processor **214** may load the next available sample block. In an embodiment, the sample blocks may be assembled according to the operations of method **400** described above with reference to FIG. **4**. In an embodiment, the next available sample block may be an oldest in time sample block which has not been analyzed to determine whether signals of interest are present in the sample block. In block **508** the processor **214** may average the amplitude measurements in the sample block. In determination block **510** the processor **214** may determine whether the average for the sample block is greater than or equal to the noise floor average set in block **502**. In this manner, sample blocks including potential signals may be quickly distinguished from sample blocks which may not include potential signals reducing processing time by enabling sample blocks without potential signals to be identified and ignored. If the average for the sample block is lower than the noise floor average (i.e., determination block **510**="No"), no signals of interest may be present in the current sample block. In determination block **514** the processor **214** may determine whether a cross block flag is set. If the cross block flag is not set (i.e., determination block **514**="No"), in block **506** the processor **214** may load the next available sample block and in block **508** average the sample block **508**.

(65) If the average of the sample block is equal to or greater than the noise floor average (i.e., determination block **510**="Yes"), the sample block may potentially include a signal of interest and in block **512** the processor **214** may reset a measurement counter (C) to 1. The measurement counter value indicating which sample within a sample block is under analysis. In determination block **516** the processor **214** may determine whether the RF measurement of the next frequency sample (C) is greater than the signal power threshold. In this manner, the value of the measurement counter (C) may be used to control which sample RF measurement in the sample block is compared to the signal power threshold. As an example, when the counter (C) equals 1, the first RF measurement may be checked against the signal power threshold and when the counter (C) equals 2 the second RF measurement in the sample block may be checked, etc. If the C RF measurement is less than or equal to the signal power threshold (i.e., determination block **516**="No"), in determination block **517** the processor **214** may determine whether the cross block flag is set. If the cross block flag is not set (i.e., determination block **517**="No"), in determination block **522** the processor **214** may determine whether the end of the sample block is reached. If the end of the sample block is reached (i.e., determination block **522**="Yes"), in block **506** the processor **214** may load the next available sample block and proceed in blocks **508**, **510**, **514**, and **512** as discussed above. If the end of the sample block is not reached (i.e., determination block **522**="No"), in block **524** the processor **214** may increment the measurement counter (C) so that the next sample in the sample block is analyzed.

(66) If the C RF measurement is greater than the signal power threshold (i.e., determination block **516**="Yes"), in block **518** the processor **214** may check the status of the cross block flag to determine whether the cross block flag is set. If the cross block flag is not set (i.e., determination block **518**="No"), in block **520** the processor **214** may set a sample start. As an example, the processor **214** may set a sample start by indicating a potential signal of interest may be discovered in a memory by assigning a memory location for RF measurements associated with the sample start. Referring to FIG. **5B**, in block **526** the processor **214** may store the C RF measurement in a memory location for the sample currently under analysis. In block **528** the processor **214** may increment the measurement counter (C) value.

(67) In determination block **530** the processor **214** may determine whether the C RF measurement (e.g., the next RF measurement because the value of the RF measurement counter was incremented) is greater than the signal power threshold. If the C RF measurement is greater than the signal power threshold (i.e., determination block **530**="Yes"), in determination block **532** the processor **214** may determine whether the end of the sample block is reached. If the end of the sample block is not reached (i.e., determination block **532**="No"), there may be further RF

measurements available in the sample block and in block **526** the processor **214** may store the C RF measurement in the memory location for the sample. In block **528** the processor may increment the measurement counter (C) and in determination block **530** determine whether the C RF measurement is above the signal power threshold and in block **532** determine whether the end of the sample block is reached. In this manner, successive sample RF measurements may be checked against the signal power threshold and stored until the end of the sample block is reached and/or until a sample RF measurement falls below the signal power threshold. If the end of the sample block is reached (i.e., determination block **532**="Yes"), in block **534** the processor **214** may set the cross block flag. In an embodiment, the cross block flag may be a flag in a memory available to the processor **214** indicating the signal potential spans across two or more sample blocks. In a further embodiment, prior to setting the cross block flag in block **534**, the slope of a line drawn between the last two RF measurement samples may be used to determine whether the next sample block likely contains further potential signal samples. A negative slope may indicate that the signal of interest is fading and may indicate the last sample was the final sample of the signal of interest. In another embodiment, the slope may not be computed and the next sample block may be analyzed regardless of the slope.

(68) If the end of the sample block is reached (i.e., determination block **532**="Yes") and in block **534** the cross block flag is set, referring to FIG. 5A, in block **506** the processor **214** may load the next available sample block, in block **508** may average the sample block, and in block **510** determine whether the average of the sample block is greater than or equal to the noise floor average. If the average is equal to or greater than the noise floor average (i.e., determination block **510**="Yes"), in block **512** the processor **214** may reset the measurement counter (C) to 1. In determination block **516** the processor **214** may determine whether the C RF measurement for the current sample block is greater than the signal power threshold. If the C RF measurement is greater than the signal power threshold (i.e., determination block **516**="Yes"), in determination block **518** the processor **214** may determine whether the cross block flag is set. If the cross block flag is set (i.e., determination block **518**="Yes"), referring to FIG. 5B, in block **526** the processor **214** may store the C RF measurement in the memory location for the sample and in block **528** the processor may increment the measurement counter (C). As discussed above, in blocks **530** and **532** the processor **214** may perform operations to determine whether the C RF measurement is greater than the signal power threshold and whether the end of the sample block is reached until the C RF measurement is less than or equal to the signal power threshold (i.e., determination block **530**="No") or the end of the sample block is reached (i.e., determination block **532**="Yes"). If the end of the sample block is reached (i.e., determination block **532**="Yes"), as discussed above in block **534** the cross block flag may be set (or verified and remain set if already set) and in block **535** the C RF measurement may be stored in the sample.

(69) If the end of the sample block is reached (i.e., determination block **532**="Yes") and in block **534** the cross block flag is set, referring to FIG. 5A, the processor may perform operations of blocks **506**, **508**, **510**, **512**, **516**, and **518** as discussed above. If the average of the sample block is less than the noise floor average (i.e., determination block **510**="No") and the cross block flag is set (i.e., determination block **514**="Yes"), the C RF measurement is less than or equal to the signal power threshold (i.e., determination block **516**="No") and the cross block flag is set (i.e., determination block **517**="Yes"), or the C RF measurement is less than or equal to the signal power threshold (i.e., determination block **516**="No"), referring to FIG. 5B, in block **538** the processor **214** may set the sample stop. As an example, the processor **214** may indicate that a sample end is reached in a memory and/or that a sample is complete in a memory. In block **540** the processor **214** may compute and store complex I and Q data for the stored measurements in the sample. In block **542** the processor **214** may determine a mean of the complex I and Q data. Referring to FIG. 5C, in determination block **544** the processor **214** may determine whether the mean of the complex I and Q data is greater than a signal threshold. If the mean of the complex I and Q data is less than or

equal to the signal threshold (i.e., determination block **544**="No"), in block **550** the processor **214** may indicate the sample is noise and discard data associated with the sample from memory.

(70) If the mean is greater than the signal threshold (i.e., determination block **544**="Yes"), in block **546** the processor **214** may identify the sample as a signal of interest. In an embodiment, the processor **214** may identify the sample as a signal of interest by assigning a signal identifier to the signal, such as a signal number or sample number. In block **548** the processor **214** may determine and store signal parameters for the signal. As an example, the processor **214** may determine and store a frequency peak of the identified signal, a peak power of the identified signal, an average power of the identified signal, a signal bandwidth of the identified signal, and/or a signal duration of the identified signal. In block **552** the processor **214** may clear the cross block flag (or verify that the cross block flag is unset). In block **556** the processor **214** may determine whether the end of the sample block is reached. If the end of the sample block is not reached (i.e., determination block **556**="No") in block **558** the processor **214** may increment the measurement counter (C), and referring to FIG. 5A in determination block **516** may determine whether the C RF measurement is greater than the signal power threshold. Referring to FIG. 5C, if the end of the sample block is reached (i.e., determination block **556**="Yes"), referring to FIG. 5A, in block **506** the processor **214** may load the next available sample block.

(71) FIG. 6 illustrates a process flow for an embodiment method **600** for displaying signal identifications. In an embodiment, the operations of method **600** may be performed by a processor **214** of a spectrum management device **202**. In determination block **602** the processor **214** may determine whether a signal is identified. If a signal is not identified (i.e., determination block **602**="No"), in block **604** the processor **214** may wait for the next scan. If a signal is identified (i.e., determination block **602**="Yes"), in block **606** the processor **214** may compare the signal parameters of an identified signal to signal parameters in a history database **232**. In determination block **608** the processor **214** may determine whether signal parameters of the identified signal match signal parameters in the history database **232**. If there is no match (i.e., determination block **608**="No"), in block **610** the processor **214** may store the signal parameters as a new signal in the history database **232**. If there is a match (i.e., determination block **608**="Yes"), in block **612** the processor **214** may update the matching signal parameters as needed in the history database **232**.

(72) In block **614** the processor **214** may compare the signal parameters of the identified signal to signal parameters in a signal characteristic listing **236**. In an embodiment, the characteristic listing **236** may be a static database separate from the history database **232**, and the characteristic listing **236** may correlate signal parameters with signal identifications. In determination block **616** the processor **214** may determine whether the signal parameters of the identified signal match any signal parameters in the signal characteristic listing **236**. In an embodiment, the match in determination **616** may be a match based on a tolerance between the signal parameters of the identified signal and the parameters in the characteristic listing **236**. If there is a match (i.e., determination block **616**="Yes"), in block **618** the processor **214** may indicate a match in the history database **232** and in block **622** may display an indication of the signal identification on a display **242**. As an example, the indication of the signal identification may be a display of the radio call sign of an identified FM radio station signal. If there is not a match (i.e., determination block **616**="No"), in block **620** the processor **214** may display an indication that the signal is an unidentified signal. In this manner, the user may be notified a signal is present in the environment, but that the signal does not match to a signal in the characteristic listing.

(73) FIG. 7 illustrates a process flow of an embodiment method **700** for displaying one or more open frequency. In an embodiment, the operations of method **700** may be performed by the processor **214** of a spectrum management device **202**. In block **702** the processor **214** may determine a current location of the spectrum management device **202**. In an embodiment, the processor **214** may determine the current location of the spectrum management device **202** based on location inputs received from a location receiver **212**, such as GPS coordinates received from a

GPS receiver **212**. In block **704** the processor **214** may compare the current location to the stored location value in the historical database **232**. As discussed above, the historical or history database **232** may be a database storing information about signals previously actually identified by the spectrum management device **202**. In determination block **706** the processor **214** may determine whether there are any matches between the location information in the historical database **232** and the current location. If there are no matches (i.e., determination block **706**="No"), in block **710** the processor **214** may indicate incomplete data is available. In other words the spectrum data for the current location has not previously been recorded.

(74) If there are matches (i.e., determination block **706**="Yes"), in optional block **708** the processor **214** may display a plot of one or more of the signals matching the current location. As an example, the processor **214** may compute the average frequency over frequency intervals across a given spectrum and may display a plot of the average frequency over each interval. In block **712** the processor **214** may determine one or more open frequencies at the current location. As an example, the processor **214** may determine one or more open frequencies by determining frequency ranges in which no signals fall or at which the average is below a threshold. In block **714** the processor **214** may display an indication of one or more open frequency on a display **242** of the spectrum management device **202**.

(75) FIG. **8A** is a block diagram of a spectrum management device **802** according to an embodiment. Spectrum management device **802** is similar to spectrum management device **202** described above with reference to FIG. **2A**, except that spectrum management device **802** may include symbol module **816** and protocol module **806** enabling the spectrum management device **802** to identify the protocol and symbol information associated with an identified signal as well as protocol match module **814** to match protocol information. Additionally, the characteristic listing **236** of spectrum management device **802** may include protocol data **804**, hardware data **808**, environment data **810**, and noise data **812** and an optimization module **818** may enable the signal processor **214** to provide signal optimization parameters.

(76) The protocol module **806** may identify the communication protocol (e.g., LTE, CDMA, etc.) associated with a signal of interest. In an embodiment, the protocol module **806** may use data retrieved from the characteristic listing, such as protocol data **804** to help identify the communication protocol. The symbol detector module **816** may determine symbol timing information, such as a symbol rate for a signal of interest. The protocol module **806** and/or symbol module **816** may provide data to the comparison module **222**. The comparison module **222** may include a protocol match module **814** which may attempt to match protocol information for a signal of interest to protocol data **804** in the characteristic listing to identify a signal of interest. Additionally, the protocol module **806** and/or symbol module **816** may store data in the memory module **226** and/or history database **232**. In an embodiment, the protocol module **806** and/or symbol module **816** may use protocol data **804** and/or other data from the characteristic listing **236** to help identify protocols and/or symbol information in signals of interest.

(77) The optimization module **818** may gather information from the characteristic listing, such as noise figure parameters, antenna hardware parameters, and environmental parameters correlated with an identified signal of interest to calculate a degradation value for the identified signal of interest. The optimization module **818** may further control the display **242** to output degradation data enabling a user of the spectrum management device **802** to optimize a signal of interest.

(78) FIG. **8B** is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment. Only those logical operations illustrated in FIG. **8B** different from those described above with reference to FIG. **2B** will be discussed. As illustrated in FIG. **8B**, as received time tracking **850** may be applied to the I and Q data from the receiver **210**. An additional buffer **851** may further store the I and Q data received and a symbol detector **852** may identify the symbols of a signal of interest and determine the symbol rate. A multiple access scheme identifier module **854** may identify whether a signal is

part of a multiple access scheme (e.g., CDMA), and a protocol identifier module **856** may attempt to identify the protocol the signal of interest is associated with. The multiple access scheme identifier module **854** and protocol identifier module **856** may retrieve data from the static database **238** to aid in the identification of the access scheme and/or protocol. The symbol detector module **852** may pass data to the signal parameter and protocol module **858** which may store protocol and symbol information in addition to signal parameter information for signals of interest.

(79) FIG. **9** illustrates a process flow of an embodiment method **900** for determining protocol data and symbol timing data. In an embodiment, the operations of method **900** may be performed by the processor **214** of a spectrum management device **802**. In determination block **902** the processor **214** may determine whether two or more signals are detected. If two or more signals are not detected (i.e., determination block **902**="No"), in determination block **902** the processor **214** may continue to determine whether two or more signals are detected. If two or more signals are detected (i.e., determination block **902**="Yes"), in determination block **904** the processor **214** may determine whether the two or more signals are interrelated. In an embodiment, a mean correlation value of the spectral decomposition of each signal may indicate the two or more signals are interrelated. As an example, a mean correlation of each signal may generate a value between 0.0 and 1, and the processor **214** may compare the mean correlation value to a threshold, such as a threshold of 0.75. In such an example, a mean correlation value at or above the threshold may indicate the signals are interrelated while a mean correlation value below the threshold may indicate the signals are not interrelated and may be different signals. In an embodiment, the mean correlation value may be generated by running a full energy bandwidth correlation of each signal, measuring the values of signal transition for each signal, and for each signal transition running a spectral correlation between signals to generate the mean correlation value. If the signals are not interrelated (i.e., determination block **904**="No"), the signals may be two or more different signals, and in block **907** processor **214** may measure the interference between the two or more signals. In an optional embodiment, in optional block **909** the processor **214** may generate a conflict alarm indicating the two or more different signals interfere. In an embodiment, the conflict alarm may be sent to the history database and/or a display. In determination block **902** the processor **214** may continue to determine whether two or more signals are detected. If the two signal are interrelated (i.e., determination block **904**="Yes"), in block **905** the processor **214** may identify the two or more signals as a single signal. In block **906** the processor **214** may combine signal data for the two or more signals into a signal single entry in the history database. In determination block **908** the processor **214** may determine whether the signals mean averages. If the mean averages (i.e., determination block **908**="Yes"), the processor **214** may identify the signal as having multiple channels in determination block **910**. If the mean does not average (i.e., determination block **908**="No") or after identifying the signal as having multiple channels in determination block **910**, in block **914** the processor **214** may determine and store protocol data for the signal. In block **916** the processor **214** may determine and store symbol timing data for the signal, and the method **900** may return to block **902**.

(80) FIG. **10** illustrates a process flow of an embodiment method **1000** for calculating signal degradation data. In an embodiment, the operations of method **1000** may be performed by the processor **214** of a spectrum management device **202**. In block **1002** the processor may detect a signal. In block **1004** the processor **214** may match the signal to a signal in a static database. In block **1006** the processor **214** may determine noise figure parameters based on data in the static database **236** associated with the signal. As an example, the processor **214** may determine the noise figure of the signal based on parameters of a transmitter outputting the signal according to the static database **236**. In block **1008** the processor **214** may determine hardware parameters associated with the signal in the static database **236**. As an example, the processor **214** may determine hardware parameters such as antenna position, power settings, antenna type, orientation, azimuth, location, gain, and equivalent isotropically radiated power (EIRP) for the transmitter associated with the

signal from the static database **236**. In block **1010** processor **214** may determine environment parameters associated with the signal in the static database **236**. As an example, the processor **214** may determine environment parameters such as rain, fog, and/or haze based on a delta correction factor table stored in the static database and a provided precipitation rate (e.g., mm/hr). In block **1012** the processor **214** may calculate and store signal degradation data for the detected signal based at least in part on the noise figure parameters, hardware parameters, and environmental parameters. As an example, based on the noise figure parameters, hardware parameters, and environmental parameters free space losses of the signal may be determined. In block **1014** the processor **214** may display the degradation data on a display **242** of the spectrum management device **202**. In a further embodiment, the degradation data may be used with measured terrain data of geographic locations stored in the static database to perform pattern distortion, generate propagation and/or next neighbor interference models, determine interference variables, and perform best fit modeling to aide in signal and/or system optimization.

(81) FIG. **11** illustrates a process flow of an embodiment method **1100** for displaying signal and protocol identification information. In an embodiment, the operations of method **1100** may be performed by a processor **214** of a spectrum management device **202**. In block **1102** the processor **214** may compare the signal parameters and protocol data of an identified signal to signal parameters and protocol data in a history database **232**. In an embodiment, a history database **232** may be a database storing signal parameters and protocol data for previously identified signals. In block **1104** the processor **214** may determine whether there is a match between the signal parameters and protocol data of the identified signal and the signal parameters and protocol data in the history database **232**. If there is not a match (i.e., determination block **1104**="No"), in block **1106** the processor **214** may store the signal parameters and protocol data as a new signal in the history database **232**. If there is a match (i.e., determination block **1104**="Yes"), in block **1108** the processor **214** may update the matching signal parameters and protocol data as needed in the history database **232**.

(82) In block **1110** the processor **214** may compare the signal parameters and protocol data of the identified signal to signal parameters and protocol data in the signal characteristic listing **236**. In determination block **1112** the processor **214** may determine whether the signal parameters and protocol data of the identified signal match any signal parameters and protocol data in the signal characteristic listing **236**. If there is a match (i.e., determination block **1112**="Yes"), in block **1114** the processor **214** may indicate a match in the history database and in block **1118** may display an indication of the signal identification and protocol on a display. If there is not a match (i.e., determination block **1112**="No"), in block **1116** the processor **214** may display an indication that the signal is an unidentified signal. In this manner, the user may be notified a signal is present in the environment, but that the signal does not match to a signal in the characteristic listing.

(83) FIG. **12A** is a block diagram of a spectrum management device **1202** according to an embodiment. Spectrum management device **1202** is similar to spectrum management device **802** described above with reference to FIG. **8A**, except that spectrum management device **1202** may include TDOA/FDOA module **1204** and modulation module **1206** enabling the spectrum management device **1202** to identify the modulation type employed by a signal of interest and calculate signal origins. The modulation module **1206** may enable the signal processor to determine the modulation applied to signal, such as frequency modulation (e.g., FSK, MSK, etc.) or phase modulation (e.g., BPSK, QPSK, QAM, etc.) as well as to demodulate the signal to identify payload data carried in the signal. The modulation module **1206** may use payload data **1221** from the characteristic listing to identify the data types carried in a signal. As examples, upon demodulating a portion of the signal the payload data may enable the processor **214** to determine whether voice data, video data, and/or text based data is present in the signal. The TDOA/FDOA module **1204** may enable the signal processor **214** to determine time difference of arrival for signals of interest and/or frequency difference of arrival for signals of interest. Using the TDOA/FDOA information

estimates of the origin of a signal may be made and passed to a mapping module **1225** which may control the display **242** to output estimates of a position and/or direction of movement of a signal.

(84) FIG. **12B** is a schematic logic flow block diagram illustrating logical operations which may be performed by a spectrum management device according to an embodiment. Only those logical operations illustrated in FIG. **12B** different from those described above with reference to FIG. **8B** will be discussed. A magnitude squared **1252** operation may be performed on data from the symbol detector **852** to identify whether frequency or phase modulation is present in the signal. Phase modulated signals may be identified by the phase modulation **1254** processes and frequency modulated signals may be identified by the frequency modulation **1256** processes. The modulation information may be passed to a signal parameters, protocols, and modulation module **1258**.

(85) FIG. **13** illustrates a process flow of an embodiment method **1300** for estimating a signal origin based on a frequency difference of arrival. In an embodiment, the operations of method **1300** may be performed by a processor **214** of a spectrum management device **1202**. In block **1302** the processor **214** may compute frequency arrivals and phase arrivals for multiple instances of an identified signal. In block **1304** the processor **214** may determine frequency difference of arrival for the identified signal based on the computed frequency difference and phase difference. In block **1306** the processor may compare the determined frequency difference of arrival for the identified signal to data associated with known emitters in the characteristic listing to estimate an identified signal origin. In block **1308** the processor **214** may indicate the estimated identified signal origin on a display of the spectrum management device. As an example, the processor **214** may overlay the estimated origin on a map displayed by the spectrum management device.

(86) FIG. **14** illustrates a process flow of an embodiment method for displaying an indication of an identified data type within a signal. In an embodiment, the operations of method **1400** may be performed by a processor **214** of a spectrum management device **1202**. In block **1402** the processor **214** may determine the signal parameters for an identified signal of interest. In block **1404** the processor **214** may determine the modulation type for the signal of interest. In block **1406** the processor **214** may determine the protocol data for the signal of interest. In block **1408** the processor **214** may determine the symbol timing for the signal of interest. In block **1410** the processor **214** may select a payload scheme based on the determined signal parameters, modulation type, protocol data, and symbol timing. As an example, the payload scheme may indicate how data is transported in a signal. For example, data in over the air television broadcasts may be transported differently than data in cellular communications and the signal parameters, modulation type, protocol data, and symbol timing may identify the applicable payload scheme to apply to the signal. In block **1412** the processor **214** may apply the selected payload scheme to identify the data type or types within the signal of interest. In this manner, the processor **214** may determine what type of data is being transported in the signal, such as voice data, video data, and/or text based data. In block **1414** the processor may store the data type or types. In block **1416** the processor **214** may display an indication of the identified data types.

(87) FIG. **15** illustrates a process flow of an embodiment method **1500** for determining modulation type, protocol data, and symbol timing data. Method **1500** is similar to method **900** described above with reference to FIG. **9**, except that modulation type may also be determined. In an embodiment, the operations of method **1500** may be performed by a processor **214** of a spectrum management device **1202**. In blocks **902**, **904**, **905**, **906**, **908**, and **910** the processor **214** may perform operations of like numbered blocks of method **900** described above with reference to FIG. **9**. In block **1502** the processor may determine and store a modulation type. As an example, a modulation type may be an indication that the signal is frequency modulated (e.g., FSK, MSK, etc.) or phase modulated (e.g., BPSK, QPSK, QAM, etc.). As discussed above, in block **914** the processor may determine and store protocol data and in block **916** the processor may determine and store timing data.

(88) In an embodiment, based on signal detection, a time tracking module, such as a TDOA/FDOA

module **1204**, may track the frequency repetition interval at which the signal is changing. The frequency repetition interval may also be tracked for a burst signal. In an embodiment, the spectrum management device may measure the signal environment and set anchors based on information stored in the historic or static database about known transmitter sources and locations. In an embodiment, the phase information about a signal be extracted using a spectral decomposition correlation equation to measure the angle of arrival (“AOA”) of the signal. In an embodiment, the processor of the spectrum management device may determine the received power as the Received Signal Strength (“RSS”) and based on the AOA and RSS may measure the frequency difference of arrival. In an embodiment, the frequency shift of the received signal may be measured and aggregated over time. In an embodiment, after an initial sample of a signal, known transmitted signals may be measured and compared to the RSS to determine frequency shift error. In an embodiment, the processor of the spectrum management device may compute a cross ambiguity function of aggregated changes in arrival time and frequency of arrival. In an additional embodiment, the processor of the spectrum management device may retrieve FFT data for a measured signal and aggregate the data to determine changes in time of arrival and frequency of arrival. In an embodiment, the signal components of change in frequency of arrival may be averaged through a Kalman filter with a weighted tap filter from 2 to 256 weights to remove measurement error such as noise, multipath interference, etc. In an embodiment, frequency difference of arrival techniques may be applied when either the emitter of the signal or the spectrum management device are moving or when then emitter of the signal and the spectrum management device are both stationary. When the emitter of the signal and the spectrum management device are both stationary the determination of the position of the emitter may be made when at least four known other known signal emitters positions are known and signal characteristics may be available. In an embodiment, a user may provide the four other known emitters and/or may use already in place known emitters, and may use the frequency, bandwidth, power, and distance values of the known emitters and their respective signals. In an embodiment, where the emitter of the signal or spectrum management device may be moving, frequency difference of arrival techniques may be performed using two known emitters.

(89) FIG. **16** illustrates an embodiment method for tracking a signal origin. In an embodiment, the operations of method **1600** may be performed by a processor **214** of a spectrum management device **1202**. In block **1602** the processor **214** may determine a time difference of arrival for a signal of interest. In block **1604** the processor **214** may determine a frequency difference of arrival for the signal interest. As an example, the processor **214** may take the inverse of the time difference of arrival to determine the frequency difference of arrival of the signal of interest. In block **1606** the processor **214** may identify the location. As an example, the processor **214** may determine the location based on coordinates provided from a GPS receiver. In determination block **1608** the processor **214** may determine whether there are at least four known emitters present in the identified location. As an example, the processor **214** may compare the geographic coordinates for the identified location to a static database and/or historical database to determine whether at least four known signals are within an area associated with the geographic coordinates. If at least four known emitters are present (i.e., determination block **1608**=“Yes”), in block **1612** the processor **214** may collect and measure the RSS of the known emitters and the signal of interest. As an example, the processor **214** may use the frequency, bandwidth, power, and distance values of the known emitters and their respective signals and the signal of interest. If less than four known emitters are present (i.e., determination block **1608**=“No”), in block **1610** the processor **214** may measure the angle of arrival for the signal of interest and the known emitter. Using the RSS or angle or arrival, in block **1614** the processor **214** may measure the frequency shift and in block **1616** the processor **214** may obtain the cross ambiguity function. In determination block **1618** the processor **214** may determine whether the cross ambiguity function converges to a solution. If the cross ambiguity function does converge to a solution (i.e., determination block **1618**=“Yes”), in block **1620** the

processor **214** may aggregate the frequency shift data. In block **1622** the processor **214** may apply one or more filter to the aggregated data, such as a Kalman filter. Additionally, the processor **214** may apply equations, such as weighted least squares equations and maximum likelihood equations, and additional filters, such as a non-line-of-sight (“NLOS”) filters to the aggregated data. In an embodiment, the cross ambiguity function may resolve the position of the emitter of the signal of interest to within 3 meters. If the cross ambiguity function does not converge to a solution (i.e., determination block **1618**=“No”), in block **1624** the processor **214** may determine the time difference of arrival for the signal and in block **1626** the processor **214** may aggregate the time shift data. Additionally, the processor may filter the data to reduce interference. Whether based on frequency difference of arrival or time difference of arrival, the aggregated and filtered data may indicate a position of the emitter of the signal of interest, and in block **1628** the processor **214** may output the tracking information for the position of the emitter of the signal of interest to a display of the spectrum management device and/or the historical database. In an additional embodiment, location of emitters, time and duration of transmission at a location may be stored in the history database such that historical information may be used to perform and predict movement of signal transmission. In a further embodiment, the environmental factors may be considered to further reduce the measured error and generate a more accurate measurement of the location of the emitter of the signal of interest.

(90) The processor **214** of spectrum management devices **202**, **802** and **1202** may be any programmable microprocessor, microcomputer or multiple processor chip or chips that can be configured by software instructions (applications) to perform a variety of functions, including the functions of the various embodiments described above. In some devices, multiple processors may be provided, such as one processor dedicated to wireless communication functions and one processor dedicated to running other applications. Typically, software applications may be stored in the internal memory **226** or **230** before they are accessed and loaded into the processor **214**. The processor **214** may include internal memory sufficient to store the application software instructions. In many devices the internal memory may be a volatile or nonvolatile memory, such as flash memory, or a mixture of both. For the purposes of this description, a general reference to memory refers to memory accessible by the processor **214** including internal memory or removable memory plugged into the device and memory within the processor **214** itself.

(91) The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the order of steps in the foregoing embodiments may be performed in any order. Words such as “thereafter,” “then,” “next,” etc. are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Further, any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

(92) The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

(93) The hardware used to implement the various illustrative logics, logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an application

specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but, in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Alternatively, some steps or methods may be performed by circuitry that is specific to a given function.

(94) In one or more exemplary aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable medium or non-transitory processor-readable medium. The steps of a method or algorithm disclosed herein may be embodied in a processor-executable software module which may reside on a non-transitory computer-readable or processor-readable storage medium. Non-transitory computer-readable or processor-readable storage media may be any storage media that may be accessed by a computer or a processor. By way of example but not limitation, such non-transitory computer-readable or processor-readable media may include RAM, ROM, EEPROM, FLASH memory, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of non-transitory computer-readable and processor-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

(95) Identifying Devices in White Space.

(96) The present invention provides for systems, methods, and apparatus solutions for device sensing in white space, which improves upon the prior art by identifying sources of signal emission by automatically detecting signals and creating unique signal profiles. Device sensing has an important function and applications in military and other intelligence sectors, where identifying the emitter device is crucial for monitoring and surveillance, including specific emitter identification (SEI).

(97) At least two key functions are provided by the present invention: signal isolation and device sensing. Signal Isolation according to the present invention is a process whereby a signal is detected, isolated through filtering and amplification, amongst other methods, and key characteristics extracted. Device Sensing according to the present invention is a process whereby the detected signals are matched to a device through comparison to device signal profiles and may include applying a confidence level and/or rating to the signal-profile matching. Further, device sensing covers technologies that permit storage of profile comparisons such that future matching can be done with increased efficiency and/or accuracy. The present invention systems, methods, and apparatus are constructed and configured functionally to identify any signal emitting device, including by way of example and not limitation, a radio, a cell phone, etc.

(98) Regarding signal isolation, the following functions are included in the present invention: amplifying, filtering, detecting signals through energy detection, waveform-based, spectral correlation-based, radio identification-based, or matched filter method, identifying interference, identifying environmental baseline(s), and/or identify signal characteristics.

(99) Regarding device sensing, the following functions are included in the present invention: using

signal profiling and/or comparison with known database(s) and previously recorded profile(s), identifying the expected device or emitter, stating the level of confidence for the identification, and/or storing profiling and sensing information for improved algorithms and matching. In preferred embodiments of the present invention, the identification of the at least one signal emitting device is accurate to a predetermined degree of confidence between about 80 and about 95 percent, and more preferably between about 80 and about 100 percent. The confidence level or degree of confidence is based upon the amount of matching measured data compared with historical data and/or reference data for predetermined frequency and other characteristics.

(100) The present invention provides for wireless signal-emitting device sensing in the white space based upon a measured signal, and considers the basis of license(s) provided in at least one reference database, preferably the federal communication commission (FCC) and/or other defined database including license listings. The methods include the steps of providing a device for measuring characteristics of signals from signal emitting devices in a spectrum associated with wireless communications, the characteristics of the measured data from the signal emitting devices including frequency, power, bandwidth, duration, modulation, and combinations thereof; making an assessment or categorization on analog and/or digital signal(s); determining the best fit based on frequency if the measured power spectrum is designated in historical and/or reference data, including but not limited to the FCC or other database(s) for select frequency ranges; determining analog or digital, based on power and sideband combined with frequency allocation; determining a TDM/FDM/CDM signal, based on duration and bandwidth; determining best modulation fit for the desired signal, if the bandwidth and duration match the signal database(s); adding modulation identification to the database; listing possible modulations with best percentage fit, based on the power, bandwidth, frequency, duration, database allocation, and combinations thereof; and identifying at least one signal emitting device from the composite results of the foregoing steps. Additionally, the present invention provides that the phase measurement of the signal is calculated between the difference of the end frequency of the bandwidth and the peak center frequency and the start frequency of the bandwidth and the peak center frequency to get a better measurement of the sideband drop off rate of the signal to help determine the modulation of the signal.

(101) In embodiments of the present invention, an apparatus is provided for automatically identifying devices in a spectrum, the apparatus including a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device. The characteristics of signals and measured data from the signal emitting devices include frequency, power, bandwidth, duration, modulation, and combinations thereof.

(102) The present invention systems including at least one apparatus, wherein the at least one apparatus is operable for network-based communication with at least one server computer including a database, and/or with at least one other apparatus, but does not require a connection to the at least one server computer to be operable for identifying signal emitting devices; wherein each of the apparatus is operable for identifying signal emitting devices including: a housing, at least one processor and memory, and sensors constructed and configured for sensing and measuring wireless communications signals from signal emitting devices in a spectrum associated with wireless communications; and wherein the apparatus is operable to automatically analyze the measured data to identify at least one signal emitting device in near real-time from attempted detection and identification of the at least one signal emitting device.

(103) Identifying Open Space in a Wireless Communication Spectrum

(104) The present invention provides for systems, methods, and apparatus solutions for automatically identifying open space, including open space in the white space of a wireless

communication spectrum. Importantly, the present invention identifies the open space as the space that is unused and/or seldom used (and identifies the owner of the licenses for the seldom used space, if applicable), including unlicensed spectrum, white space, guard bands, and combinations thereof. Method steps of the present invention include: automatically obtaining a listing or report of all frequencies in the frequency range; plotting a line and/or graph chart showing power and bandwidth activity; setting frequencies based on a frequency step and/or resolution so that only user-defined frequencies are plotted; generating files, such as by way of example and not limitation, .csv or .pdf files, showing average and/or aggregated values of power, bandwidth and frequency for each derived frequency step; and showing an activity report over time, over day vs. night, over frequency bands if more than one, in white space if requested, in Industrial, Scientific, and Medical (ISM) band or space if requested; and if frequency space is seldom in that area, then identify and list frequencies and license holders.

(105) Additional steps include: automatically scanning the frequency span, wherein a default scan includes a frequency span between about 54 MHz and about 804 MHz; an ISM scan between about 900 MHz and about 2.5 GHz; an ISM scan between about 5 GHz and about 5.8 GHz; and/or a frequency range based upon inputs provided by a user. Also, method steps include scanning for an allotted amount of time between a minimum of about 15 minutes up to about 30 days; preferably scanning for allotted times selected from the following: a minimum of about 15 minutes; about 30 minutes; about 1 hour increments; about 5 hour increments; about 10 hour increments; about 24 hours; about 1 day; and about up to 30 days; and combinations thereof. In preferred embodiments, if the apparatus is configured for automatically scanning for more than about 15 minutes, then the apparatus is preferably set for updating results, including updating graphs and/or reports for an approximately equal amount of time (e.g., every 15 minutes).

(106) The systems, methods, and apparatus also provide for automatically calculating a percent activity associated with the identified open space on predetermined frequencies and/or ISM bands.

(107) Automated Reports and Visualization of Analytics

(108) Various reports for describing and illustrating with visualization the data and analysis of the device, system and method results from spectrum management activities include at least reports on power usage, RF survey, and variance.

(109) The systems, methods, and devices of the various embodiments enable spectrum management by identifying, classifying, and cataloging signals of interest based on radio frequency measurements. In an embodiment, signals and the parameters of the signals may be identified and indications of available frequencies may be presented to a user. In another embodiment, the protocols of signals may also be identified. In a further embodiment, the modulation of signals, devices or device types emitting signals, data types carried by the signals, and estimated signal origins may be identified.

(110) Referring again to the drawings, FIG. 17 is a schematic diagram illustrating an embodiment for scanning and finding open space. A plurality of nodes are in wireless or wired communication with a software defined radio, which receives information concerning open channels following real-time scanning and access to external database frequency information.

(111) FIG. 18 is a diagram of an embodiment of the invention wherein software defined radio nodes are in wireless or wired communication with a master transmitter and device sensing master.

(112) FIG. 19 is a process flow diagram of an embodiment method of temporally dividing up data into intervals for power usage analysis and comparison. The data intervals are initially set to seconds, minutes, hours, days and weeks, but can be adjusted to account for varying time periods (e.g., if an overall interval of data is only a week, the data interval divisions would not be weeks). In one embodiment, the interval slicing of data is used to produce power variance information and reports.

(113) FIG. 20 is a flow diagram illustrating an embodiment wherein frequency to license matching occurs. In such an embodiment the center frequency and bandwidth criteria can be checked against

a database to check for a license match. Both licensed and unlicensed bands can be checked against the frequencies, and, if necessary, non-correlating factors can be marked when a frequency is uncorrelated.

(114) FIG. **21** is a flow diagram illustrating an embodiment method for reporting power usage information, including locational data, data broken down by time intervals, frequency and power usage information per band, average power distribution, propagation models, atmospheric factors, which is capable of being represented graphical, quantitatively, qualitatively, and overlaid onto a geographic or topographic map.

(115) FIG. **22** is a flow diagram illustrating an embodiment method for creating frequency arrays. For each initialization, an embodiment of the invention will determine a center frequency, bandwidth, peak power, noise floor level, resolution bandwidth, power and date/time. Start and end frequencies are calculated using the bandwidth and center frequency and like frequencies are aggregated and sorted in order to produce a set of frequency arrays matching power measurements captured in each band.

(116) FIG. **23** is a flow diagram illustrating an embodiment method for reframe and aggregating power when producing frequency arrays.

(117) FIG. **24** is a flow diagram illustrating an embodiment method of reporting license expirations by accessing static or FCC databases.

(118) FIG. **25** is a flow diagram illustrating an embodiment method of reporting frequency power use in graphical, chart, or report format, with the option of adding frequencies from FCC or other databases.

(119) FIG. **26** is a flow diagram illustrating an embodiment method of connecting devices. After acquiring a GPS location, static and FCC databases are accessed to update license information, if available. A frequency scan will find open spaces and detect interferences and/or collisions. Based on the master device ID, set a random generated token to select channel from available channel model and continually transmit ID channel token. If node device reads ID, it will set itself to channel based on token and device will connect to master device. Master device will then set frequency and bandwidth channel. For each device connected to master, a frequency, bandwidth, and time slot in which to transmit is set. In one embodiment, these steps can be repeated until the max number of devices is connected. As new devices are connected, the device list is updated with channel model and the device is set as active. Disconnected devices are set as inactive. If collision occurs, update channel model and get new token channel. Active scans will search for new or lost devices and update devices list, channel model, and status accordingly. Channel model IDs are actively sent out for new or lost devices.

(120) FIG. **27** is a flow diagram illustrating an embodiment method of addressing collisions.

(121) FIG. **28** is a schematic diagram of an embodiment of the invention illustrating a virtualized computing network and a plurality of distributed devices. FIG. **28** is a schematic diagram of one embodiment of the present invention, illustrating components of a cloud-based computing system and network for distributed communication therewith by mobile communication devices. FIG. **28** illustrates an exemplary virtualized computing system for embodiments of the present invention loyalty and rewards platform. As illustrated in FIG. **28**, a basic schematic of some of the key components of a virtualized computing (or cloud-based) system according to the present invention is shown. The system **2800** comprises at least one remote server computer **2810** with a processing unit **2811** and memory. The server **2810** is constructed, configured and coupled to enable communication over a network **2850**. The server provides for user interconnection with the server over the network with the at least one apparatus as described hereinabove **2840** positioned remotely from the server. Furthermore, the system is operable for a multiplicity of devices or apparatus embodiments **2860**, **2870** for example, in a client/server architecture, as shown. Alternatively, interconnection through the network **2850** using the at least one device or apparatus for measuring signal emitting devices, each of the at least one apparatus is operable for network-based

communication. Also, alternative architectures may be used instead of the client/server architecture. For example, a computer communications network, or other suitable architecture may be used. The network **2850** may be the Internet, an intranet, or any other network suitable for searching, obtaining, and/or using information and/or communications. The system of the present invention further includes an operating system **2812** installed and running on the at least one remote server **2810**, enabling the server **2810** to communicate through network **2850** with the remote, distributed devices or apparatus embodiments as described hereinabove. The operating system may be any operating system known in the art that is suitable for network communication.

(122) Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

Claims

1. An apparatus for identifying open space in an electromagnetic spectrum comprising: at least one processor; and at least one memory; wherein the apparatus is operable to receive signal data of at least one signal measured by a sensor; and wherein the apparatus is operable to calculate a percent activity associated with open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the received signal data.
2. The apparatus of claim 1, wherein the open space is determined based on a dynamically adjustable interval.
3. The apparatus of claim 1, wherein identification of the open space is automated.
4. The apparatus of claim 1, wherein identification of the open space is based upon the predetermined frequencies, the user-defined frequencies, and/or the ISM bands.
5. The apparatus of claim 1, wherein identification of the open space occurs in near-real time or real time.
6. The apparatus of claim 1, wherein the apparatus is further configured to detect interference in the electromagnetic spectrum based on the signal data.
7. An apparatus for identifying open space in an electromagnetic spectrum comprising: at least one processor; and at least one memory; wherein the apparatus receives measured signal data of at least one signal; wherein the measured signal data includes energy measurements of the at least one signal, at least one parameter of the at least one signal, and/or at least one change for the at least one signal; and wherein the apparatus is operable to calculate a percent activity associated with at least one open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the measured signal data.
8. The apparatus of claim 7, wherein the apparatus is further configured to detect interference in the electromagnetic spectrum based on the measured signal data.
9. The apparatus of claim 7, wherein identification of the at least one open space occurs in near-real time or real time and/or is automated.
10. The apparatus of claim 7, wherein the at least one open space is determined based on a dynamically adjustable interval.
11. A method for identifying open space in an electromagnetic spectrum comprising: at least one apparatus including a processor and a memory receiving signal data from at least one remote sensor over a network or the at least one apparatus creating the signal data using at least one integrated sensor; and calculating a percent activity associated with at least one open space on predetermined frequencies, user-defined frequencies, and/or Industrial, Scientific, and Medical (ISM) bands based on the signal data; wherein the signal data includes energy measurements of at least one signal, at

- least one parameter of the at least one signal, and/or at least one change for the at least one signal.
12. The method of claim 11, wherein the signal data is generated by the at least one apparatus sensing and measuring signals transmitted from at least one signal emitting device in the electromagnetic spectrum.
 13. The method of claim 11 wherein identification of the at least one open space is automated.
 14. The method of claim 11 wherein the at least one open space is determined based on a dynamically adjustable interval.
 15. The method of claim 11, wherein identification of the at least one open space is based upon the predetermined frequencies, the user-defined frequencies, and/or the ISM bands.
 16. The method of claim 11, wherein identification of the at least one open space is displayed or visually indicated, in whole or in part, by the at least one apparatus.
 17. The method of claim 11, wherein identification of the at least one open space occurs in near-real time or real time.
 18. The method of claim 11, further comprising detecting interference in the electromagnetic spectrum based on the signal data.
 19. The method of claim 11, wherein the at least one open space includes unused space or unlicensed spectrum.
 20. The method of claim 11, wherein the at least one open space includes white space.
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